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Generative adversarial networks and hyperparameter-optimized XGBoost for enhanced heart disease prediction

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Heart Disease (HD) serves as the primary reason for worldwide deaths which creates an urgent requirement to create diagnostic intelligent systems that can identify patients at their earliest stage. The development of strong predictive models faces a significant obstacle because clinical data contains both biased information and noisy elements which create unreliable results. This study proposed a hybrid framework, GAN-XO, for early HD detection. The methodology addressed fundamental data quality issues through a multi-step preprocessing pipeline. First, a Generative Adversarial Network (GAN) is employed for synthetic oversampling of the minority class, balancing the imbalanced PKIOHD and Framingham datasets. Second, a two-phase outlier detection and elimination process using z-score and Interquartile Range (IQR) methods is applied to obtain clean data. Finally, an XGBoost classifier, whose hyperparameters are optimally tuned using Optuna, is utilized for prediction. The designed GAN-XO system achieved its best accuracy score of 96.60% and F1-Score value of 0.9649 through testing on an evenly distributed and completely clean test dataset. The ablation research demonstrated that all system elements provided distinct benefits while the outlier elimination process turned out to be vital for improving model performance and trustworthiness. The research results demonstrate that accurate healthcare predictions require both integrated modelling systems and precise management of data quality. The substantial performance enhancement shows that the framework improves clinical decision-making while enabling correct diagnosis of HD.

Keywords Heart disease, XGBoost, Optuna optimizer, Generative adversarial networks

As per the World Health Organization (WHO), the cardiovascular disease (CVDs) category is the major global reason responsible for human mortality[1]. HD and stroke together have main behavioural risk factors which include diets lacking health, inactivity, tobacco usage and alcohol misuse and leading environmental risk factor among others is air pollution. People can avoid developing most CVDs by controlling behavioural and environmental risk factors. Medical experts need to detect cardiovascular illnesses at an early stage to start proper treatment through prescription drugs and team counselling sessions. To address this challenge, the detection of CVDs primarily utilizes machine learning (ML) techniques to perform automated, data-driven diagnosis of CVDs. The combination of ML with Deep Learning (DL) methods improves diagnosis methods for CVDs. However, Gradient boosting (GB) as a decision tree technique has become widely used across various fields in present times[2]. Gradient Boosting Decision Tree (GBDT) achieved highly effective performance when solving classification problems[3]. Extreme Gradient Boosting (XGBoost) represents the new form of the GB algorithm, which is its most recent version[4]. The approach uses a better learning objective approximation that uses goal estimation and regularization techniques to reduce overfitting problems. The algorithm of the new leaf node estimation and the tree node splitting works as XGBoost[5]. Conversely, an effective prediction system must have an effective hyperparameter adaptive process to gain performance gains. A choice of the best hyperparameters

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will result in the substantial enhancement of the final output. The chosen option to use the hyperparameter optimization framework is the Optuna since it is a highly rated tool of optimization that brings out the best outcomes. The problem is to identify the best hyperparameter choice, also referred to as hyperparameter optimization[6,7]. The main process of HD prediction is found in Fig. 1.

In imbalanced datasets, the learning algorithms tend to focus more on the majority less than minority class. Since oversampling and undersampling methods aim to balance classes, they encounter significant issues, such as losing valuable examples or overfitting. SMOTE, a synthetic data generation technique, introduces synthetic minority examples but does not consider the distribution of the majority class, resulting in noisy examples that disturb the class boundary. SMOTE with AdaBoost can improve minority class accuracy, but only up to a point where the class becomes robust to noise and outliers[8]. Existing studies show a tendency to handle imbalance and noise problems separately through basic oversampling methods which fail to provide an effective solution. The study demonstrates that data cleansing procedures result in better model interpretability which functions as an essential requirement for clinical implementation.

To overcome these issues, a GAN was implemented, which handles the large and complex data[9]. The use of GANs cannot eliminate all challenges with generated data due to outlier problems. To address this challenge, when analyzing data with outliers, researchers rely on two statistical methods, known as the z-score and IQR, to identify outliers for elimination purposes, thereby making the remaining data more accurate[10]. The present research gap exists because current medical data processing algorithms yield inadequate results, as imbalanced datasets often fail to consider instances from minority classes. Current approaches that combine oversampling with undersampling or manual labelling often result in data misrepresentation or fail to accurately fit the data, and both methods are time-intensive and costly. The motivation of this study is to develop a solution that combines GANs with outliers and without outlier detection algorithms to automate general prediction models, thereby improving accuracy beyond current methods. The key contributions of this research study.

- **End-to-end data preprocessing**: Most current procedures only focus on a single component, whereas the proposed system is the first to offer a hybrid of GAN-based data balancing, as well as dual outlier sampling, that would guarantee a high quality of synthetic data and also impact on the model performances with the clean and outlier data.
- **State-of-the-art optimization**: Finely tuning hyperparameters of a GAN with balanced data that is generated by a GAN provides a synergistic effect, which reduces model performance better than conventional balancing methods (SMOTE and ADASYN) that have been used before, and this GAN in the HD dataset is mostly used for missing values or data augmentation.
- **Validated on challenging dataset**: The model achieves a better result under the PKIOHD dataset, which has only been used by few studies, and the proposed model has an improvement of 1.9% compared to the previous best result.
- **Better performance**: The proposed GAN-XO model has the best accuracy of 96.6% and F1-Score of 96 which is better than all the existing methods. In this study, it is rigorously revealed that outliers affect predictions and explanations when parallel SHAP and LIME analyses of contaminated and cleaned data are conducted, and the conclusions show that only after the removal of outliers does the proposed model have the top, stable, and clinically coherent performance.

In comparison to the offered GAN-XO framework, hyOPTXg utilizes data balancing through SMOTE as well as traditional hyperparameter optimization but does not use GAN-based data augmentation to improve the model performance. The use of SMOTE is a static and popular resampling approach that fills in the problem of class imbalance but might introduce unwanted and noisy samples or extreme samples unintentionally into the training set. Equally, ASEB uses SMOTE + GAN to balance the classes and does not directly deal with outliers or optimize the model parameters but only works with ensemble structures and weights. Contrarily, GAN-XO

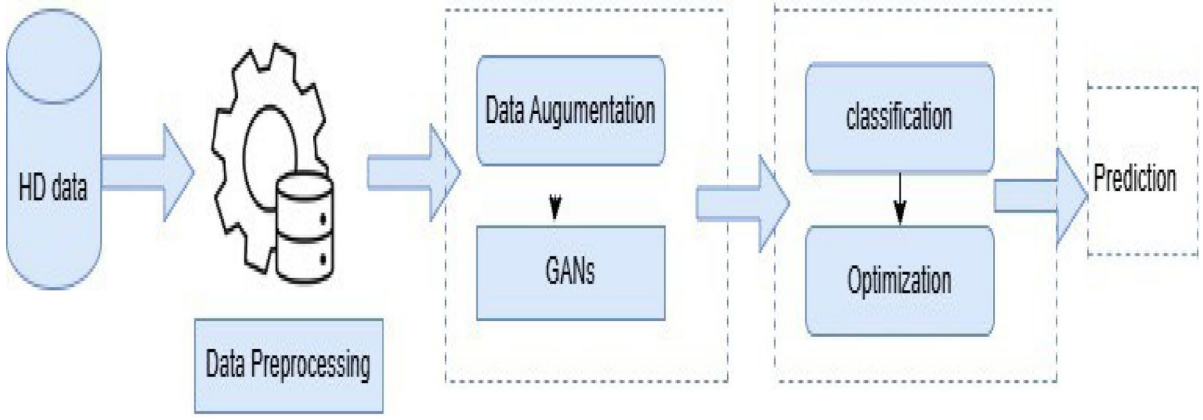


Fig. 1. Prediction Framework of the Proposed Model.

combines automatic outliers' reduction and advanced GAN-based data synthesis with refined model learning in a single pipeline, which allows learning more robustly and reliably with complex data distributions.

Literature review

Literature demonstrates exploration of HD prediction using ML as a well-established field, illustrated in Table 1. In this research work, the authors[11]employed Fast Correlation-Based Filter (FCBF), Minimum Redundancy Maximum Relevance (mRMR), and Relief, along with Particle Swarm Optimization (PSO), to choose the optimal features. The researchers utilized ML using GB, Random Forest (RF), Extra Trees (ET) Classifier, and Logistic Regression (LR) on feature-selected sets from both small and large datasets. The ET and RF models achieved the most satisfactory results because the chosen features were of high quality. The mRMR and FCBF, together with Relief, achieved complete accuracy on small data sets. ET together with RF yielded superior performance on the large dataset, while FCBF along with Relief attained 78% and 77% success rates, respectively. The research introduced[12]Attribute GAN (AGAN), which solves HD data with missing attributes by applying GANs. The implementation of an attribute matrix enables AGAN to identify correct missing value categories, so data consistency improves alongside prediction accuracy enhancement. The model adds real and generated data divergences to its loss function which enables stable and realistic data distribution computation. The performance evaluation of AGAN utilizes both Root Mean Squared Error (RMSE) and the Proportion of Falsely Classified data to indicate superior results compared to current methods.

The researchers[13]developed hyOPTXg as an optimized XGBoost ML model that functions for HD prediction through the application of optuna for hyperparameter optimization. Three datasets of Cleveland, Kaggle Heart Failure Prediction, and UCI HD produced hyOPTXg accuracy results of 94.7, 89.3, and 88.5. The multiple measures of performance evaluation prove that hyOPTXg has superior results, compared to alternative approaches, as it reaches high recall values and precision levels and F1-Score and accuracy of results with an excellent Area Under Curve-Receiver Operating Characteristic (AUC-ROC) scores. This study has created an XGBoost model to predict HD, including algorithms of z-score and IQR outliers removal, and optimizing the model hyperparameters with the help of optuna. The highest results were achieved when testing on the cleveland HD data was done in a train test ratio of 90: 10, resulting in an accuracy of 95.45, a precision of 100, and an F1-Score of 96.3. The model reliability was established by Stratified K-Fold Cross-Validation (CV) processes[14].

Most existing research focuses on combining multiple models, selecting important features, performing data processing, and utilizing SMOTE, oversampling, and undersampling techniques to address class imbalance[15]. This research utilizes GANs to detect complex patterns and interconnections within HD information. The 90:10 train-test split selection delivered the most precise results which resulted in dependable assessment according to earlier evaluations[14]. The researchers used GANs on the main dataset because this method allowed them to investigate complex relationships present in cardiac feature information. The researchers applied multiple outlier removal methods to enhance the quality of the data.

Materials and methods

Dataset cleaning and balancing

Research utilizes the PKIOHD dataset from the Kaggle Notebook: Personal Key Indicators of Heart Diseases. The dataset features have been classified into three broad types of risk factors: demographic, behavioural, and

References	Author	Year	Summary	Research Gap
12	Xing Chen et al.	2024	HD prediction based on AGAN in cases	The AGAN model architecture successfully handles data imputation tasks but fails to capture complete non-linear relational patterns of clinical risk factors, which restricts its ability to make accurate predictions across different patient groups.
13	Srinivas et al.	2022	XGBoost optimization using Optuna	The study utilized a dataset with a restricted number of clinical features, employing a modelling technique that did not capitalize on advanced feature engineering or selection methods to increase prediction accuracy.
14	Dhanka and S. Maini	2025	XGBoost with and without optimization using Optuna, tested with and without outliers across different train-test splits for HD detection.	The model's development and validation process used a dataset that contained only a restricted set of clinical variables, which limits its ability to predict outcomes in different patient groups.
16	Jiayuan Zhu et al.	2025	CVDs detection using SVM-Recursive Feature Selection cross-validation (RFECV).	While utilizing ECG/PCG data, prior models demonstrated a performance ceiling (e.g., 91.6% accuracy), highlighting an opportunity for improved frameworks to achieve higher diagnostic precision.
17	Manikandan et al.	2025	LR with and without Boruta FS for HD prediction.	Existing models for cardiac disease prediction, as reported in the literature, show room for improvement in key performance metrics such as accuracy and F1-Score on standard benchmark datasets.
18	Balasubramaniam et al.	2024	A combination of ReliefF-based FS and the Gradient Squirrel Search Algorithm (GSSA) enable the Deep MaxOut Network to detect HD.	The results show that the optimization strategy and feature selection mechanism used in this study resulted in suboptimal model performance because they produced models with reduced accuracy and slower training convergence.
19	Mohammad Alshraideh et al.	2024	HD prediction using SVM classifier optimized with PSO.	The methodology failed to include effective data quality assessment methods which handle both data imbalance and outlier detection together with automatic hyperparameter tuning procedures which either result in model performance validation.
20	Mana Saleh Al Reshan et al.	2023	HD prediction system using Hybrid Deep Neural Networks (DNN).	The method failed to use deep ensemble learning architectures and it did not perform proper model tuning for the ensemble components along with their meta-parameter settings.

Table 1. Summary of HD Prediction with Previously Proposed ML Methods.

clinical. This classification helps explain how various factors contribute to an individual's risk of HD. To collect this information, a survey was conducted, and the survey information was obtained through phone interviews with adults by CDC's Behavioural Risk Factor Surveillance System (BRFSS) in 2020. CDC collected data from over 400,000 adults in the United States. The health survey is composed primarily of binary (yes/no) questions. The cleaning process resulted in 320,000 remaining records containing 18 essential health indicators; most of the data is imbalanced. To address the issue of class imbalance, this research work employs GAN data balancing techniques, which involve generating synthetic data and expanding the dataset to 584,844 samples while retaining the original 17 features and 1 is target whether the person has HD or not. Information about the data is depicted Fig. 2.

The Framingham heart study data in Kaggle is composed of 16 attributes, 1 target variable, and 15 input features to predict HD[28,37]. GAN balancing also boosts the sample size to 7,192 records instead of 4,240 records and eliminates outliers to 5670 records, making the data more reliable, as shown in the dataset information provided in Table 2. The PKIOHD and Framingham data sets include both mixed data types which are: continuous data, binary, nominal, and ordinal data. The outlier detection was only used with continuous numeric characteristics (e.g., BMI, Physical Health, Mental Health, Sleep Time, blood pressure, glucose). Binary and categorical variables (e.g. Smoking, Sex, Stroke, Diabetic, etc.) were not included in outlier detection because z-score and IQR are not applicable to encoded categorical variables https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset.

Conditional GAN for data balancing

GANs correct the datasets' balance by generating synthetic instances for minorities[12]and also assist in filling missing values by learning patterns from existing data[20]. GANs are effective for large datasets as they capture complex distributions and improve data quality. The efficiency and scalability of GANs achieves better results when they combine high-performance computing with distributed training to process large datasets. The neural network consists of two parts which use Exponential Linear Unit (ELU) for their middle layers and Sigmoid activation for their final layers. The Generator uses random noise to generate synthetic data through its three hidden layers which contain 256 and 512 and 256 neurons. The discriminator uses three hidden layers with 256 and 128 and 64 neuron connections to analyze both real and synthetic data inputs for classification purposes. The training process consists of 15,000 epochs, utilizing the Adam optimizer with a learning rate of 0.000005 and a batch size of 128.

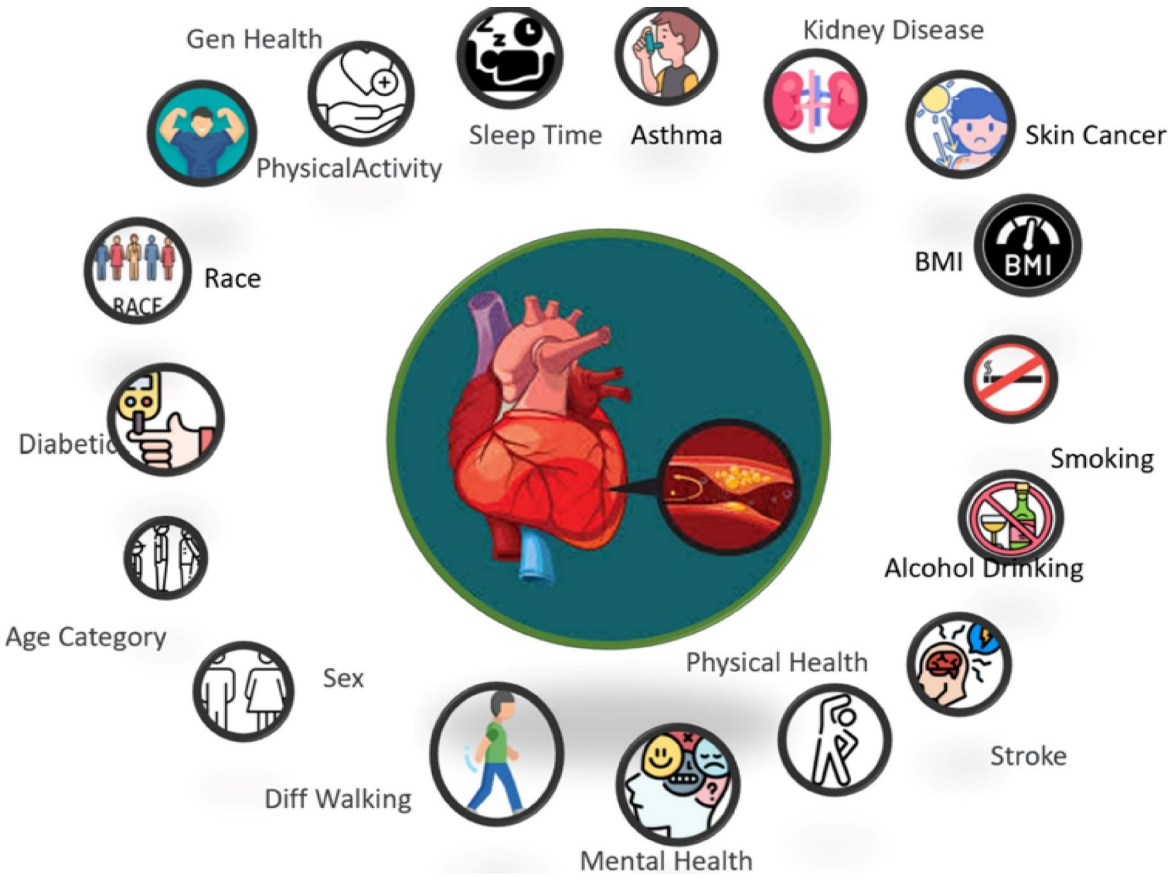


Fig. 2. Dataset Visualization.

S. no.	(a) PKIOHD data features and explanation	Variables
Demographic factors		
1	**Age Category**: Classification into different age groups.	Ordinal Categorical
2	**Sex**: Gender of the respondent.	Categorical
3	**Race**: Ethnic or racial background.	Nominal Categorical
Behavioral risk factors		
4	**Smoking**: History of smoking at least 100 cigarettes in a lifetime.	Categorical
5	**Alcohol Drinking**: Regular consumption of large amounts of alcohol.	Categorical
6	**Physical Activity**: Frequency of engaging in physical activities outside work.	Categorical
Clinical factors		
7	**BMI**: Body Mass Index, calculated using height and weight.	Categorical
8	**Diabetic**: History of diabetes diagnosis.	Categorical
9	**Gen Health**: Self-reported overall health status.	Ordinal Categorical
10	**Physical Health (PH)**: No. of days in the past month with poor physical health.	Continuous
11	**Mental Health (MH)**: No. of days in the past month with poor mental health.	Continuous
12	**Diff Walking**: Difficulty walking or climbing stairs.	Continuous
13	**Stroke**: History of being diagnosed with a stroke.	Continuous
14	**Sleep Time**: Average no. of hours slept per day.	Continuous
15	**Asthma**: History of being diagnosed with asthma.	Continuous
16	**Kidney Disease**: kidney disease history, excluding minor conditions.	Continuous
17	**Skin Cancer**: History of being diagnosed with skin cancer.	Continuous
S. no.	**(b) Framingham data feature and explanation**	**Variables**
Demographic factors		
1	**Age**: Age measured in years.	Continuous
2	**Sex**: Respondent's gender	Continuous
3	**Education**: Educational attainment level of every person.	Ordinal Categorical
Behavioural risk factors		
4	**Current Smoker**: Indicates whether the individual is a current smoker.	Categorical
5	**Cigarettes Per Day**: Average number of cigarettes smoked per day.	Categorical
Clinical factors		
6	**Body Mass Index (BMI)**: BMI is calculated using height and weight.	Categorical
7	**Blood Pressure Medication (BPMeds)**: Indicates whether the individual is taking medication for blood pressure	Categorical
8	**Prevalent Hypertension**: History of being diagnosed with hypertension.	Categorical
9	**Diabetes**: History of diabetes diagnosis.	Categorical
10	**Total Cholesterol**: Total cholesterol level measured in mg/dL.	Categorical
11	**Systolic Blood Pressure (SBP)**: Measurement in mmHg.	Categorical
12	**Diastolic Blood Pressure**: Measurement in mmHg.	Categorical
13	**Heart Rate**: Resting heart rate measured in beats per minute.	Categorical
14	**Glucose**: Blood glucose level measured in mg/dL.	Categorical
15	**Prevalent Stroke**: History of being diagnosed with a stroke.	Categorical

Table 2. Grouped PKIOHD and Framingham Dataset Features Based on Risk Factor Types. Bold values represents the general terminologies.

The researchers selected Conditional GAN (cGAN) together with its three-layer perceptron framework (Generator: 256-512-256; Discriminator: 256-128-64) because they wanted to achieve maximum performance through minimal computational requirements while handling their specific tabular medical data. The PKIOHD dataset showed through our experiments that a properly configured basic cGAN model produced synthetic samples with equal or better quality than advanced CTGAN and VAE models because it maintained both feature correlations and marginal distributions intact (Sect. 3.1.2). The selected layer dimensions were determined through a structured search, where this configuration optimally captured the non-linear relationships within our 17-dimensional feature space without exhibiting signs of underfitting or overfitting. The progressively expanding then contracting bottleneck in the Generator (256→512→256) proved effective in learning and reconstructing the complex, multi-modal distributions of clinical risk factors. Furthermore, this architecture trained stably with our chosen micro-step learning rate (5×10^{-6}), avoiding mode collapse common in GANs trained on heterogeneous data. For this study's objective, generating class-balanced, clinically plausible tabular data, the simplicity, direct controllability, and lower computational overhead of the cGAN provided a pragmatic and effective solution. The Generator Loss evaluates how effectively the generator reconstructs real data, and the Discriminator Loss checks its performance in differentiating synthetic and actual data inputs, illustrated in Fig. 3. To ensure the generated

samples belong to the target minority class, the authors employed a Conditional GAN (cGAN) architecture. The class label (heart disease yes/no (target)) was provided as an additional input to both the Generator and Discriminator, guiding the synthesis to produce clinically coherent samples specific to the underrepresented class.

After we used GAN-based oversampling, the resulting synthetic data showed biologically impossible extreme values which included BMI measurements that exceeded all human permanent limits. This phenomenon represents a recognized drawback that occurs when GAN systems train on clinical tabular datasets. We implemented a second outlier detection procedure to remove synthetic outliers from GAN-augmented datasets because these synthetic outliers would disrupt the classifier's ability to predict outcomes. The procedure eliminated approximately ~ 455,300 records which exceeded acceptable limits according to z-score and IQR standards thus establishing a final training dataset that included only medically acceptable records. The researchers selected a low learning rate of ($\eta = 5 \times 10^{-6}$) because they wanted to maintain stable adversarial training through this value which applies to both generator and discriminator models in clinical tabular data that contains numerous dimensions and different data types. In preliminary experiments, higher learning rates (e.g., 10^{-3} to 10^{-4}) led to rapid divergence of the discriminator loss, causing the generator to fail to converge. The adopted micro-step learning rate allowed for gradual weight updates which prevented one network from overpowering the other while the GAN learned to model the complex joint distributions of demographic data together with behavioral data and clinical data throughout 15,000 training epochs.

The generator produces only a small number of samples because of mode collapse which is a known danger that affects GANs during the generation of tabular data which contains both discrete and continuous features. To address this problem the authors used a cGAN architecture which allowed us to provide class labels as extra input for both the generator and the discriminator. The conditioning process directs synthesis to create class-specific feature distributions which results in diverse outcomes for every class. The training process of both networks was monitored through loss curve analysis which we conducted throughout the training process to verify that neither loss reached a zero value which indicates training collapse. The system achieved stable training results because the generator and discriminator losses maintained equal balance while the Adam optimizer adjusted learning rates for each parameter which resulted in the production of clinically plausible synthetic samples that experienced only mild mode collapse. Finally, the clinical consistency validation in Sect. 3.1.2, comparing correlation structures between real and generated data, served as an external check for mode diversity, confirming that the GAN learned the full data distribution rather than a subset of modes. The combination of these strategies, conditional training, micro-step learning, and continuous monitoring, resulted in stable adversarial training without signs of mode collapse. This is empirically supported by the preserved correlation structures and marginal distributions between real and synthetic data shown in Sect. 3.1.2, confirming that the GAN learned a diverse and clinically plausible representation of the minority class rather than replicating a few memorized samples.

The PKIOHD and Framingham datasets comprise mixed data types, continuous (e.g., BMI), binary (e.g., Smoking), and ordinal/categorical (e.g., Age Category, GenHealth). To process these for GAN training, all

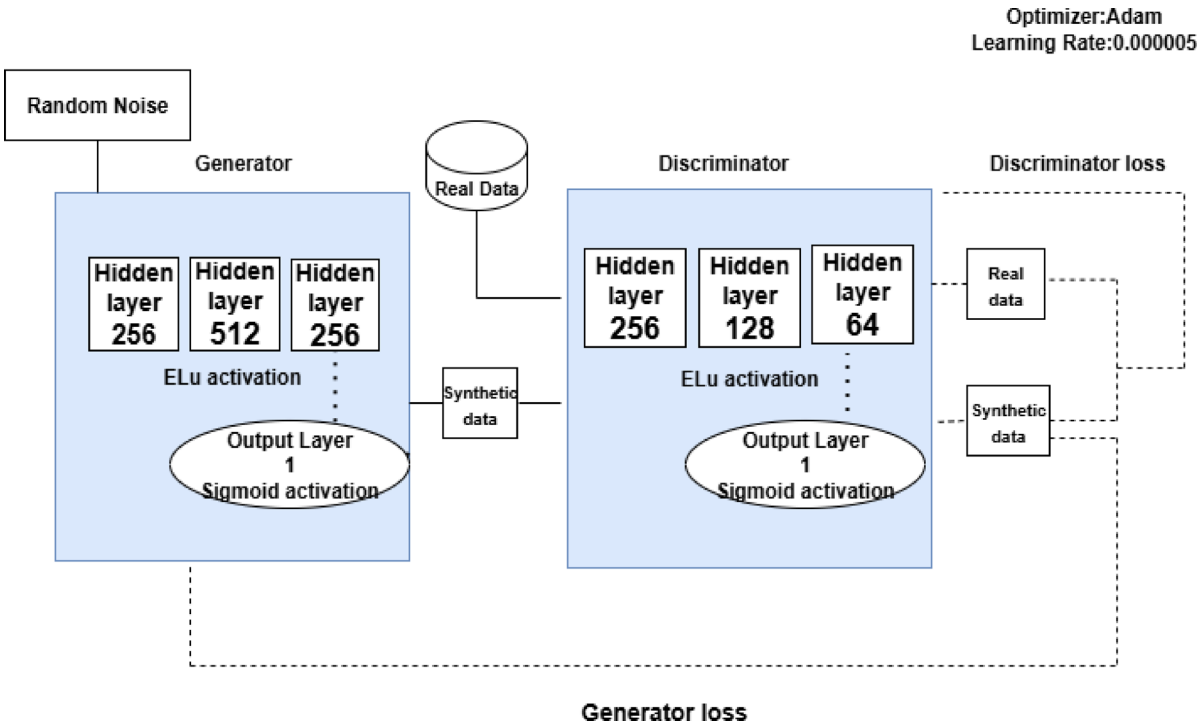


Fig. 3. GANs Procedure.

categorical features were one-hot encoded. Binary features were represented as single binary columns (0/1). This encoding scheme transforms the heterogeneous tabular data into a unified numerical representation suitable for neural network processing. The conditional GAN (cGAN) architecture was selected to provide explicit control over the class of generated samples. The generator G takes as input a noise vector $\mathrm{z} \sim \mathcal{N}(0,1)$ concatenated with a conditional vector c representing the target HD class label. The discriminator D receives both real and synthetic data which it processes together with the condition c. The conditioning process directs the synthesis process to target minority class feature distributions which enhances both training stability and sample relevance. The generator produces vectors which exist within the continuous space of the encoded features during inference. The post-processing step creates valid synthetic samples through a one-hot encoded categorical output which converts to discrete values by using an argmax function across each one-hot group. Binary feature outputs use a threshold of 0.5 for output determination. The system uses continuous outputs exactly as they are because they already show scaled numerical values. The pipeline requires all synthetic records to follow the discrete-continuous pattern found in the original clinical data.

The selection of a custom cGAN model was made because it provided better results than dedicated table-based models which included CTGAN and TVAE. Our main objective required us to create stable synthesis methods for class balancing instead of achieving precise distributional replication which CTGAN provides through its advanced mixed data modeling capabilities. The PKIOHD data subset testing showed that a properly configured cGAN model could produce authentic clinical samples which enhanced classifier accuracy while requiring less processing power and setup work. The use of a micro-step learning rate ($\eta = 5 \times 10^{-6}$), ELU activations for robust gradient flow, and the conditional framework provided sufficient control to learn the salient patterns in the data without the need for more complex, tailor-made losses. The clinical consistency of the generated data, validated in Sect. 3.1.2, confirms the adequacy of this approach for the task of augmentation.

From Kaggle, the dataset consists of 319,795 samples based on 17 features that involve HD only. After balancing, the dataset consisted of 584,844 records, and a preliminary outlier removal step was performed to eliminate extreme and anomalous values that could bias the model's performance. A GAN was employed to address the issue of class imbalance and enhance the representation of minority classes. Secondly, an additional set of more comprehensive outlier removal was executed, removing an additional 455,300 outliers. A total of 129,544 high-quality records were obtained from the dataset, making it suitable for further analysis. In order to guarantee the clinical plausibility of GAN-generated samples, statistical validation was conducted through comparing the feature-wise distributions and correlation patterns of actual and artificial data. The samples generated did not distort the ranges of the original data or the relationship between the features and no combination of features that was clinically impossible was introduced. The GAN augmentation was thus applied to enhance the balance of the classes without altering the real medical nature.

While GAN-based augmentation effectively addresses class imbalance, it introduces inherent uncertainties that must be acknowledged. Synthetic data generation creates distributional shifts and mode collapse artifacts and synthetic outliers which disrupt the stability of downstream classifiers. The uncertainties of this study relate to two main problems in probabilistic risk assessment which require precise management of aleatory uncertainties and epistemic uncertainties that arise from inherent randomness and knowledge limitations respectively 32. Uncertainties in medical decision-making research have major importance because they determine how models are calibrated and how decision boundaries stay stable and how trustworthy clinical outcomes become. The dual outlier removal strategy serves not only to clean the data but also to mitigate uncertainty propagation from synthetic samples to the XGBoost classifier, enhancing the robustness of the final predictive model.

Validation of clinical consistency in GAN-Generated data

To evaluate the clinical plausibility of the samples generated by GAN, Checked whether the synthetic data is able to retain the real clinical inter-feature relationships, as opposed to the artificial ones. Compared correlation structures over original and GAN generated datasets of both lifestyle-based and clinical datasets. In the case of the PKIOHD dataset illustrated in Fig. 4(a), correlation analysis was only done on continuous numerical variables, since yes/no (target) numbers are not applicable in the Pearson correlation analysis. In case with Framingham data, all the variables are numerical or even binary-coded, and it is possible to perform a complete correlation analysis. The resulting heatmaps indicate that the known clinical relationships, including systolic and diastolic blood pressure, smoking status and cigarette intake, diabetes and blood glucose, age and blood pressure are always maintained following the augmentation of GAN illustrated in Fig. 4(b). Even though there are insignificant changes in the strength of the correlations, the direction and the overall pattern of associations are consistent, which suggests that the GAN-created data is clinically sound and does not yield unrealistic associations of medical attributes.

The preserved correlation structures demonstrate that the GAN learned the underlying clinical relationships within the data rather than generating arbitrary feature combinations. This directly addresses the concern regarding the clinical plausibility of synthetic tabular data. While simpler methods like SMOTE interpolate between existing minority samples, they can produce unrealistic samples in high-dimensional space and ignore feature correlations. The analysis proves that the GAN-based method creates synthetic patients who exhibit both statistical validity and clinical authenticity which adds valuable distinctiveness to the training data instead of creating duplicate data points.

Detected and removed outliers

The study used both z-scores and IQR methods to identify and remove outliers from the data[10]. The researchers used the calculated z-scores to assess how points were distributed between the mean value and their respective upper and lower bounds. The z-score calculation process follows the procedure described in Eq. (1).

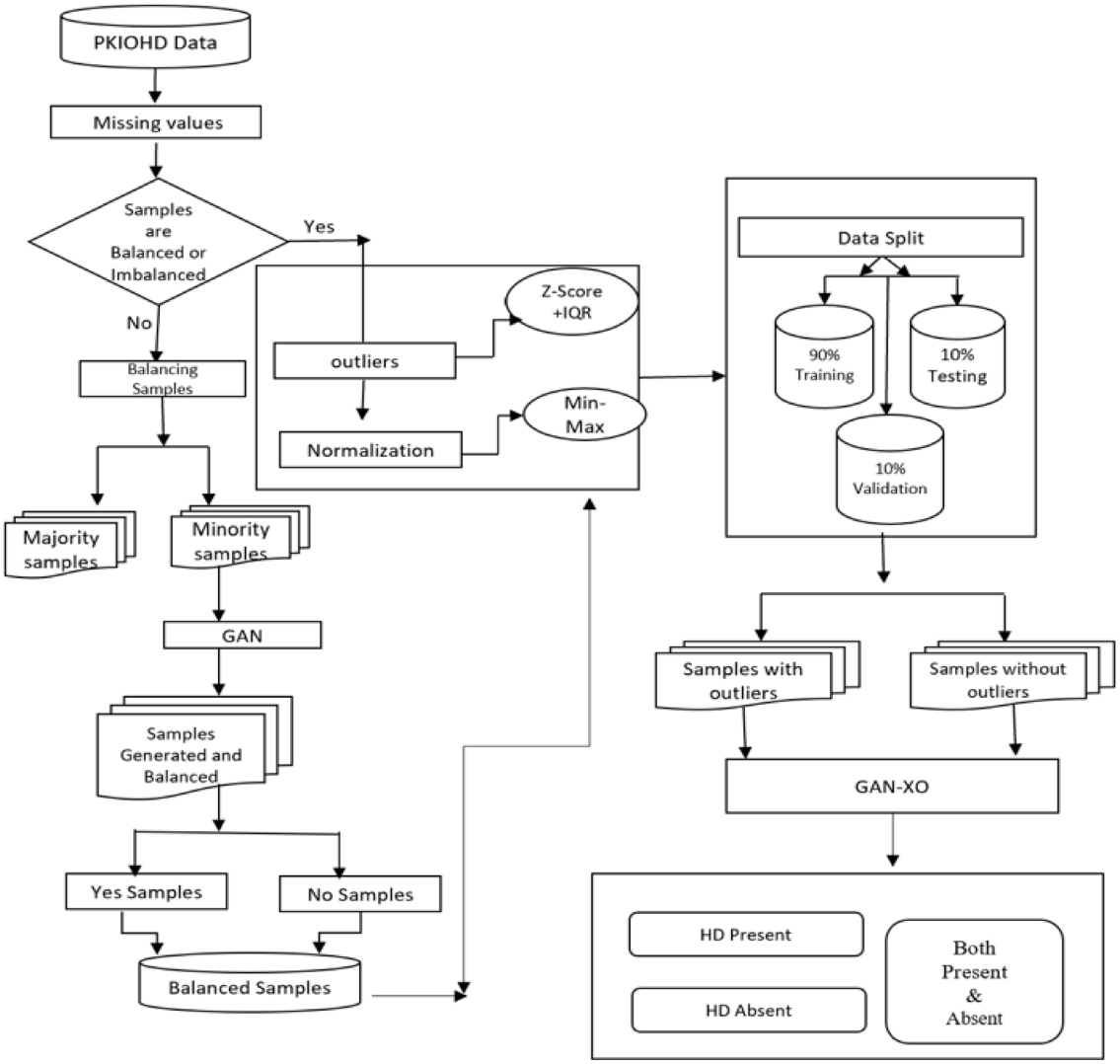


Fig. 4. Correlation Heatmap Comparison between Original and GAN-Generated Datasets for both PKIOHD and Framingham Datasets.

$$Z = \frac{(x - \mu)}{\sigma} \tag{1}$$

Outliers are detected when a data point's z-score exceeds a value of three, indicating it is far from the mean. This methodology enabled the removal of such data points from the dataset. Where:

- x represents the observation value,
- μ is the mean,
- σ is the standard deviation.

Aimed at evaluating the data distribution and assessing which values are considerably different in respect to the rest of the numbers. To prevent data leakage, μ and σ were estimated solely from the training set and then applied to the validation and test sets. Observations with $|\ z\ | > 3$ were flagged as outliers. Subtraction between the third quartile (TQ) and the first quartile (FQ) gives IQR. The value that is less than 25% of the data points is represented by the first quartile (FQ) and likewise, the third quartile (TQ) presents the value that is below 75% of the data points. Identification of outliers is done by finding out whether any data points necessarily fall outside the lower and upper limits that are determined as shown in the following Eqs. (2) and (3):

$$\text{Lower Bound (LB)} = \text{FQ} - 1.5 \times \text{IQR} \tag{2}$$

$$\text{Upper Bound (UB)} = \text{TQ} + 1.5 \times \text{IQR} \tag{3}$$

Any data that fall outside of these boundaries, whether higher than the UB or lower than the LB, are considered outliers and are typically removed from the dataset. A data distribution was carried out prior to data processing as well as after data processing to ensure good data quality and model validity. Z-score and IQR thresholds were used to perform outlier detection in an unbiased way of labeling. Removed samples were analyzed and found both the heart disease and non-heart disease records were equally distributed, which implies that there were no cases of high-risk cases that were selectively removed. This step was aimed at enhancing the stability of the models and not to filter out clinically meaningful patients. Outliers are not to be eliminated in categorical variables but only in continuous numerical variables. The thresholds were chosen based on established clinical plausibility ranges:; BMI: Retained records within 15–50 kg/m² (extremes < 15 or > 50 likely indicate measurement or data entry errors). PhysicalHealth/MentalHealth: Days in the past month capped at 30; values > 30 are impossible and indicate data corruption. SleepTime: Retained records between 3 and 14 h per day; values outside this range are physiologically improbable for adults. Age (Framingham): Plausible range 30–90 years for a cardiovascular study.

These thresholds were applied using the IQR method (1.5×IQR) and z-score (|z|>3) concurrently. Records flagged by either method were removed. The high number of removals (~ 455,300) is attributed to (a) the large size of the GAN-augmented dataset, and (b) the removal of synthetic samples that fell outside clinically credible ranges. A class-balance check confirmed that the proportion of heart disease cases in the removed set (8.5%) matched the original dataset (8.7%), indicating no systematic bias against the minority class. For instance, an example of a removed 'outlier' was a record with a PhysicalHealth value of 30 (indicating poor health every day of the previous month) combined with a BMI of 14 (severely underweight), yet labeled as 'No Heart Disease.' While theoretically possible, such extreme and contradictory combinations are likely data entry errors or rare, non-representative cases that would act as high-leverage noise during model training.

The authors acknowledge that aggressive, label-agnostic outlier removal in medical data risks discarding legitimate high-risk patients. To investigate potential selection bias, the authors analyzed the class label distribution within the removed 455,300 outliers from the PKIOHD dataset. The proportion of HD cases in the removed set (approximately 8.5%) was found to be consistent with the original pre-balanced dataset's HD class ratio (approximately 8.7%), indicating no systematic exclusion of the minority class. This analysis suggests the removed points were likely measurement errors, data entry anomalies, or extreme but non-pathological physiological values, rather than clinically critical cases of HD. Therefore, this rigorous cleansing step was designed to enhance model stability and generalization by removing points that could act as noisy, high-leverage samples during XGBoost training, without introducing a discriminatory bias against the target condition. It was an intentional decision to use the aggressive dual-filter outlier removal method (z-score = $|z| \geq 3$ and 1.5 IQR) because GAN-augmented synthetic data are known to introduce an artifact in such a large-scale synthetic dataset. GAN-based samples may trap mode collapse, distributional shift or implausible extreme values and increase the noise in the sequential boosting of XGBoost resulting in unstable decision boundaries and overfitting to synthetic outliers. To evaluate the possible bias, a more conservative, label-agnostic approach was chosen over more moderate approaches (e.g., domain-specific truncation of BMI to 10–70 or SleepTime to 3–14) to guarantee high-fidelity training data and avoid a priori assumptions about clinical constraints on all features.To evaluate possible bias, the distribution of classes in the dropped 45,5300 records was examined, the fraction of HD (positive) cases was about 8.5% which closely corresponds with the ratio of the pre-balanced original dataset of about this implies that there is no systematic lock-out of the minority category and that the deleted points are mainly measurement error, entry marbles or non-pathological synthetic extremes as opposed to clinically critical high-risk profiles. The samples (removed) were also equally spread across classes, which is in favor of contributing to the objective of improving models stability without any discriminatory effect.

Normalization

Normalization is a method used to adjust the scale of data, ensuring that values fit within a specific range. This technique ensures that comparisons between different features are valid. To normalize a data point y, the following Eq. (4) is applied:

$$y^{*} = \frac{y - \text{MinValue}}{\text{MaxValue} - \text{MinValue}} \times (\text{UpperBound} - \text{LowerBound}) + \text{LowerBound} \tag{4}$$

In this formula, "Min Value" and "Max Value" refer to the lowest and highest values in the dataset, respectively. The "Upper Bound" and "Lower Bound" represent the target range into which the data is scaled. This process ensures that the data is transformed into the specified range while preserving the relative differences between data points[21,22].

From the Sect. 3.1 and 3.2 the primary objective of conducting the aggressive outlier removal is that this study is aimed at estimating the performance of the GAN-generated synthetic data under the dual outlier removal circumstances. The majority of literature only works with small or intermediate imbalanced data and does outlier filtering on a small scale. However, unlike them, our methodology initializes synthetic balanced data with GANs and then performs tremendous outlier removal on the exponentuated dataset. This is not artificial performance enhancement through artificial increase of the dataset size and also finding complexityand constraint, but instead adds to the computational burden and provides a realistic challenge to dual outlier removal. The study assesses the effectiveness of outlier filtering of GAN-generated samples by eliminating a significant fraction of outliers, particularly when the data being generated by a GAN has high-variance elements.

Notably, the authors are not dependent on filtered information. The authors also have outliers in a parallel configuration to be able to compare the performance without and with outliers removal. It enables us to learn the behavior of model performance in both cases and the extent of GAN-based data generation at increasing the

existence of outliers. Computationally, it is not possible to operate with large synthetic datasets. Thus, controlled dual outliers removal allows us to test: the effect of the GAN generation on outlier density, their performance with outliers, performance of models following aggressive outlier filtering. The reason why this research direction is required is that, GAN-based augmentation is not thoroughly studied on the large-scale outlier interactions, and our results give an idea of whether aggressive outlier removal should be discouraged or encouraged in the following research. The dataset partition and information are illustrated in Table 3 for the both datasets.

XGBoost

XGBoost is a boosting algorithm, which was launched in 2016[23]. It is a type of ML that can be used to make better decisions since it is a combination of several models. Being a subdivision of ensemble learning, it incorporates multiple smaller models, mainly decision trees to form stronger final model. XGBoost, unlike classical approaches, has a principle of boosting of models sequential construction, the errors of each of them refining the errors of the previous one. This repetitive form of learning will make the performance very high with time[23–26]. XGBoost has the main benefits of fastness and high computation efficiency, which makes it faster than most other traditional boosting. It also includes regularization measures to avoid overfitting to guarantee improved generalization to unobserved data. It also supports missing values, large data, and unbalanced distribution, thus its use is possible in fields like healthcare, finance, and classification. In medical practice, XGBoost has been applied in the diagnosis of HD, prediction of cancer progression and detection of lung diseases, usually with superior performance over other machine learning models[27]. In this study by the author[28], an improved diagnostic methodology was proposed, making use of XGBoost, and performed better than the classifiers such as RF and ET. Also, the methodology presented in[29]emphasizes its capacity to process sparse and weighted data, compute in parallel, and regularize which helps to minimize overfitting. The gradient boosting objective optimization function at iteration k is given as follows:

$$F^{(k)} = \sum_{m=1}^{M} \Psi\left(b_m, \tilde{b}_m^{(k-1)} + g_k\left(z_m\right)\right) + P\left(g_k\right) + D \tag{5}$$

Where:

- k is the current boosting iteration,
- g_k are the newly added function refining predictions,
- Ψ - loss function,
- b_m is the real value for the mth instance,
- $\tilde{b}_m^{(k-1)}$ is the predicted value from the previous iteration,
- $P\left(g_k\right)$ is a regularization term controlling complexity,
- m is the total no. of samples,
- D is a constant that does not influence optimization.

To simplify the optimization process, a second-order Taylor approximation is applied:

$$F^{(k)} = \sum_{m=1}^{M} \left[\Psi\left(b_m, \tilde{b}_m^{(k-1)}\right) + v_m g_k\left(z_m\right) + \frac{1}{2} w_m g_k^2\left(z_m\right)\right] + P\left(g_k\right) + D \tag{6}$$

Where:

- $v_m = \frac{\partial F}{\partial \tilde{b}_m^{(k-1)}}$ is the gradient,
- $w_m = \frac{\partial^2 F}{\partial \tilde{b}_m^{(k-1)2}}$ is the Hessian matrix.

The method combines both gradient information and Hessian information to improve optimization results which enable exact solution updates while maintaining control over model complexity. The tree-based ensemble

Stages	PKIOHD (with outliers)	PKIOHD (without outliers)	Framingham (with outliers)	Framingham (without outliers)
Initial dataset size	~400,000	~400,000	4,240	4,240
After preprocessing	~320,000	~320,000	4,240	4,240
After GAN balancing	5,84,844	5,84,844	7,192	7,192
Records before outlier removal	5,84,844	5,84,844	7,192	7,192
Records after outlier removal	5,84,844	1,29,544	7,192	5,670
Number of outliers removed	0	4,55,300	0	1,522
Training records	4,67,880	1,03,635	5,752	4,536
Validation records	58,479	12,954	720	567
Testing records	58,485	12,955	720	567

Table 3. Data Processing Pipeline Summary.

method XGBoost delivers excellent results but it displays particular uncertainty patterns which need assessment for its use in medical settings. The sequential boosting process creates higher error rates because it amplifies errors from noisy or synthetic samples through its decision path which transfers input data uncertainty. The system safety principles established that decision-making which involves uncertain situations requires systems that can provide trustworthy solutions[33]. The system uses GAN-based augmentation combined with aggressive outlier filtering to decrease uncertainty propagation because it ensures synthetic samples remain clinically plausible while extreme values which usually show high uncertainty areas get addressed through systematic methods. The approach uses advanced safety engineering methods which require AI-based systems to achieve better predictive performance while maintaining system trustworthiness[34].

Optuna

The Hyperparameter optimization of optuna recreates the search process as a global optimization problem and is a better adaptive and efficient procedure compared to the method of using Grid SearchCV and Randomized SearchCV. Optuna relies on Bayesian optimization techniques to dynamically narrow the search space according to the already evaluated outcomes, as opposed to more traditional techniques that rely on exhaustive testing or random sampling. The performance of models benefits from optuna because it handles the adjustment of hyperparameters as well as feature optimization and redundancy elimination. The optimal configuration of every ensemble model before combination produces better predictive accuracy, reduces overfitting, and enhances generalization compared to traditional tuning approaches[30]. The dynamic and optimal XGBoost hyperparameters employed in this research, for models trained with and without outliers, are detailed in Table 4. During optimization, the dynamic values represent the entire exploration area, while the best parameters indicate which configuration yields superior model performance.

Proposed GAN-XO

The XGBoost model optimization through optuna performed a parameter search on max depth, reg alpha, reg lambda, min child weight, gamma, learning rate, colsample bytree, subsample, along with objective, despite all other settings remaining at their defaults. XGBoost used binary: logistic objectives for classification together with the gbtree booster. The optimization steps followed the AUC score evaluation in stratified K-Fold CV. GANs were employed for data augmentation through synthetic data generation to handle data imbalance. A process of training and assessment was conducted with and without outlier removal from the data after implementing the z-score and IQR methods to improve data quality. A performance evaluation of the model was performed by comparing its results under two different testing scenarios. The proposed methodology is illustrated in Fig. 5 and in Algorithm 1 (Table 5).

Experimental result

The experimental system was the equipped, with an 11th Gen Intel(R) Core(TM) i5-11300 H processor at 3.10 GHz, 3110 MHz, featuring 4 cores and 8 logical processors, and ran Microsoft Windows 11 Home Single Language with 16.0 GB (15.8 GB usable) of RAM, alongside Python Version 3.8.5. During the research, utilized the NumPy (1.21.5) library, along with Pandas (1.2.2) and Matplotlib (3.3.2). Scikit-learn, XGBoost, Optuna, and TensorFlow for building and training the Generative Adversarial Network. The SHAP and LIME libraries were used for model interpretability. The Conditional GAN (cGAN) for data balancing was implemented with a generator network (256-512-256 neurons) and a discriminator (256-128-64 neurons), using ELU activations in hidden layers and sigmoid in the output layers. A critical finding during development was the need for an extremely low learning rate ($\eta = 5 \times 10^{-6}$) to ensure stable adversarial training over 15,000 epochs, avoiding the mode collapse common in vanilla GANs. The XGBoost classifier was subsequently optimized using the Optuna framework over 50 trials, with the search space detailed in Table 4. The adopted low learning rate protocol resulted in stable and convergent training dynamics. As shown in Fig. 6, both generator and discriminator losses stabilized after several thousand epochs, oscillating near the theoretical equilibrium point ($\ln(2) \approx 0.693$). This stability was crucial for generating high-fidelity synthetic samples for the minority class. Ablation tests

Hyperparameters	Dynamic values
n_estimators	50–200
max_depth	3–10
Learning_rate	0.01–0.3
subsample	0.5–1.0
Colsample_bytree	0.5–1.0
gamma	0–5
min_child_weight	1–10
reg_alpha	0–10
reg_lambda	0–10
objective	binary: logistic
booster	gbtree
base_score	0.5

Table 4. XGBoost Tuned Hyperparameters.

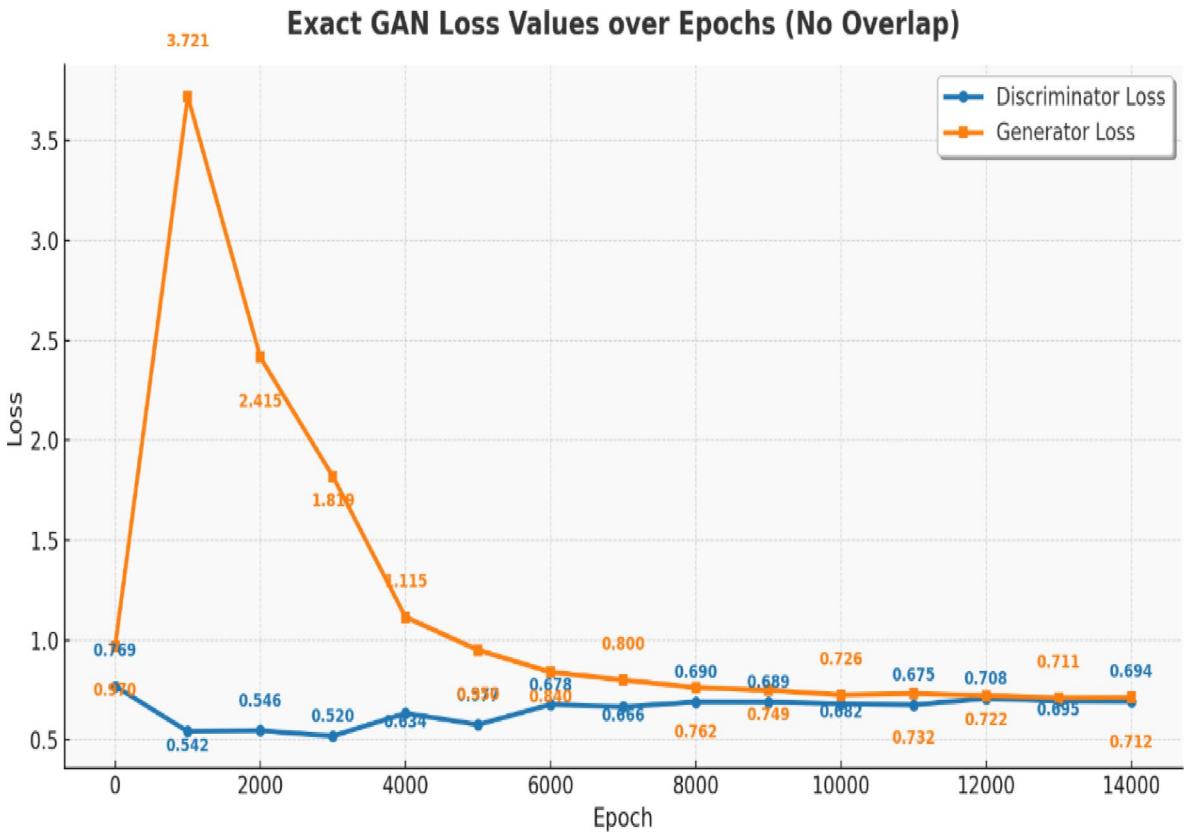


Fig. 5. Proposed Workflow.

Algorithm 1: XGBoost Hyperparameter Optimization using Optuna with GAN Balancing

1: **Step 1:** Load dataset $D = \{(x_i, y_i)\}^n$, handle missing values, normalize features
2: **Step 2:** Split into train/test sets: D_{train}, D_{test}
3: Step 3: GAN-based Data Balancing
4: Initialize Generator G [256, 512, 256] and Discriminator D [256, 128, 64] neurons
5: Set ELU activation (hidden), Sigmoid (output), epochs=15,000, learning rate=0.000005, batch=128
6: **for** epoch = 1 to 15,000 **do**
7: Train D: minimize $L_D = -\log D(x_{real}) - \log(1 - D(G(z)))$ 9:
Train G: minimize $L_G = -\log D(G(z))$ where $z \sim N(0, 1)$ 10: **end for**
8: Create balanced dataset $D_{balanced}$
9: **Step 4:** Define objective function for Optuna trials
10: **function** Objective(trial): suggest θ, train XGBoost ($D_{balanced}$, θ), return accuracy
11: **Step 5:** Run optimization: study.optimize(objective, n trials = 50)
12: **Step 6:** Train final model with best params on $D_{balanced}$
13: **Step 7:** Evaluate on D_{test}: compute accuracy, precision, recall, F1-Score.

Table 5. Proposed Algorithm for XGBoost Hyperparameter Optimization using Optuna.

confirmed that higher learning rates (e.g., 1×10^{-3}) led to training divergence, highlighting the sensitivity of GANs on this tabular medical data.

The dynamic and optimal XGBoost hyperparameters of both datasets employed in this research, for models trained with and without outliers, are detailed in Table 6, and the progress is illustrated in Fig. 7. During optimization, the dynamic values represent the entire exploration area, while the best parameters indicate which configuration yields superior model performance.

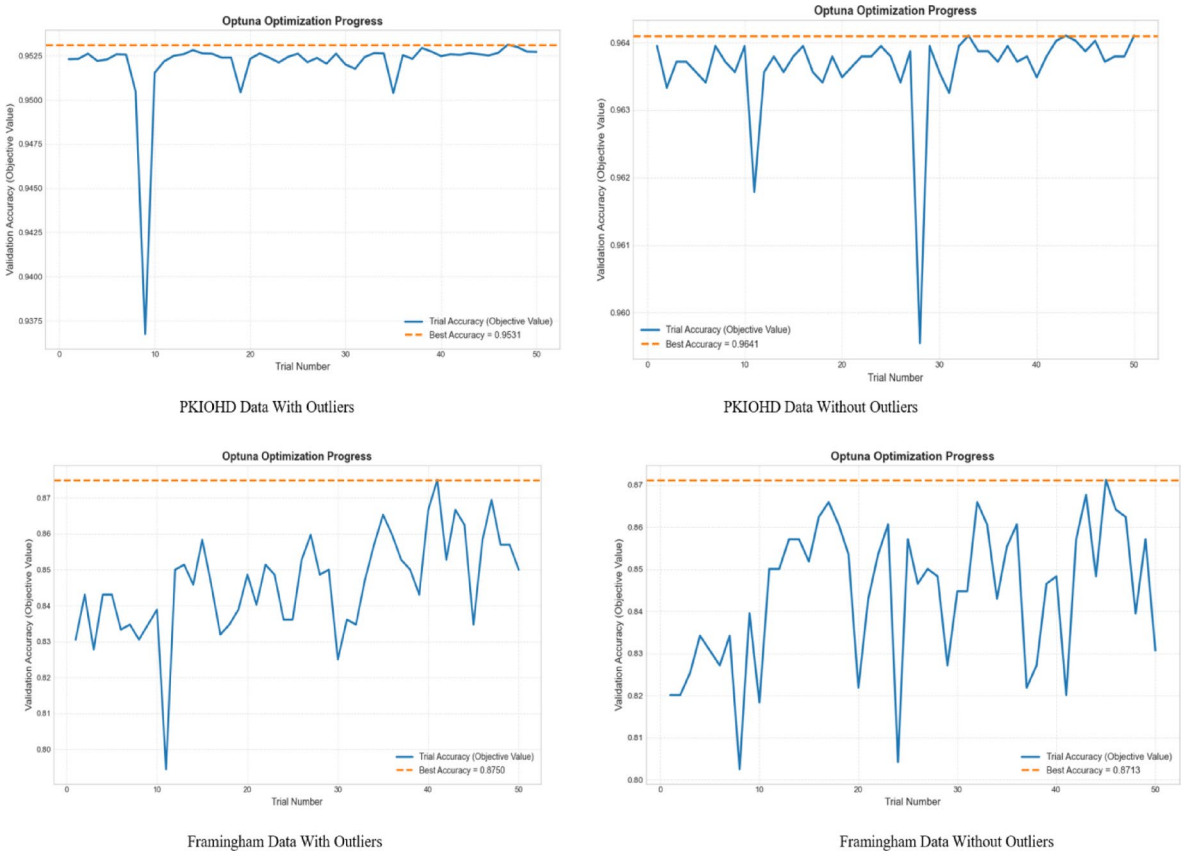


Fig. 6. Best Values for the Optuna Trails.

	PKIOHD dataset		Framingham dataset	
Hyperparameter	**Without outliers**	**With outliers**	**Without outliers**	**With outliers**
n_estimators	200	157	181	123
max_depth	8	6	8	5
Learning_rate	0.242735714	0.223332999	0.1136	0.1581
subsample	0.790605368	0.963256991	0.7723	0.7639
Colsample_bytree	0.543166028	0.517221036	0.7126	0.9571
gamma	2.947590256	1.121569926	1.1432	1.4345
min_child_weight	4.655226264	4.455961015	4.056	2.612
reg_alpha	1.566584419	1.613904314	0.9568	0.2818
reg_lambda	6.387895205	2.88717086	7.5501	0.5278
objective	binary: logistic	binary: logistic	binary: logistic	binary: logistic
booster	gbtree	gbtree	gbtree	gbtree
base_score	0.5	0.5	0.5	0.5

Table 6. Dynamically Best XGBoost Tuned Hyperparameters.

Both training stability and generalization ability were improved when the model was trained without outliers. Training Loss = 0.118561 and Validation Loss = 0.119970 were the approximate loss values after 500 epochs, suggesting a nearly perfect alignment between the two an ideal feature of a well-generalized model. By contrast, the model that was trained using outliers displayed Training Loss = 0.121717 and Validation Loss = 0.127055, which indicates a slight overfitting and a slower rate of convergence. These findings verify that eliminating outliers greatly improves the GAN's overall performance consistency and learning efficiency. A novel and enhanced type of adversarial learning called GAN-XO remains remarkably robust in the presence of outliers. It demonstrates good consistency with outliers, with training accuracy of 0.9547 and validation accuracy of 0.9531 (a negligible difference of 0.0016). Outliers are eliminated, resulting in nearly identical training accuracy of 0.9643 and validation accuracy of 0.9641 (difference of only 0.0002). This indicates that GAN-XO is a strong and trustworthy model for managing outliers in generative modeling since it performs exceptionally well in both

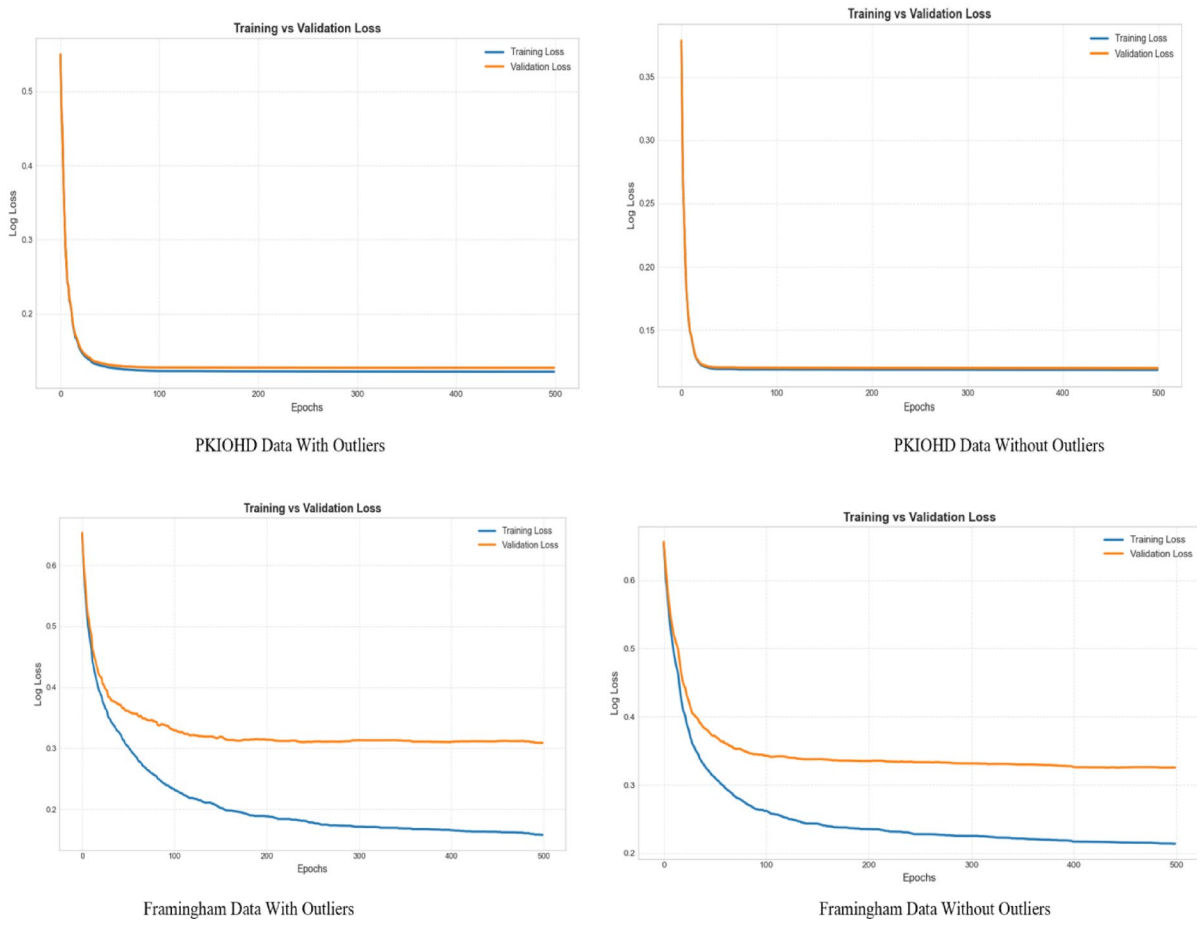


Fig. 7. Training vs. Validation Log Loss Curve.

clean and noisy data. Similarly, In the case of the Framingham data, due to the existence of outliers, there is a visible difference between the training and validation accuracy, which implies a decrease in generalization and even overfitting. Withoutlier removal leads to improved model learning and more stable training and validation curves, indicating a better generalization and the achievement of better stability, as illustrated in Figs. 8 and 9, respectively.

GAN-XO with outliers

The 90:10 Train-Test Ratio (TTR) method was implemented for data partitioning with outliers after performing data balancing through a GAN for the both dataset. The outlier dataset consisted of 584,844 records, divided into training (467,880 records), validation (58,479 records), and testing (58,485 records) sets for the PKIOHD data. And the dataset consists of 7,192 records, which are split into 5,752 training samples, 720 validation samples, and 720 testing samples for the Framingham data. This split ensures adequate data for model training, hyperparameter tuning, and unbiased performance evaluation. The performance metrics used by the authors, along with their corresponding formulas, are provided in Table 7. The entire data processing procedure through all the data steps is demonstrated in Table 8 (Algorithm 2). First, the raw data is subjected to extensive preprocessing, such as data cleaning, categorical variables encoding, class balancing through GAN and primary quality evaluation. Then, the z-score and IQR methods of outlier detection and removal are undertaken to drop intense and noisy data points and enhance the reliability of the data. Once these preprocessing activities have been done on the balanced dataset, then the refined data will be split into training and evaluation data (validation) and evaluation data (testing). This designed process is consistent in data preparation and strong in downstream learning and assessment procedures. Results from the confusion matrix indicated 28,990 True Negative (TN) along with 253 False Positive (FP), 2,475 False Negative (FN), as well as 26,767 True Positive (TP)for the PKIOHD data and framingham data is also as shown in Table 9. The model displayed 95.37% accuracy, 95.60% precision and 95.34% recall together with 95.33% F1-Score and 99.13% specificity and 0.9849 AUC-ROC score for the PKIOHD data, Simultaneously in the Framingham dataset, with the outliers performace illustrated in Table 10.

.

The findings suggest that some imbalance-handling methods are more effective in some datasets than others. To isolate the impact of the data balancing technique within our pipeline, we conducted an ablation study replacing the GAN with SMOTE (creating an otherwise identical SMOTE-XO model). As shown in Table 10,

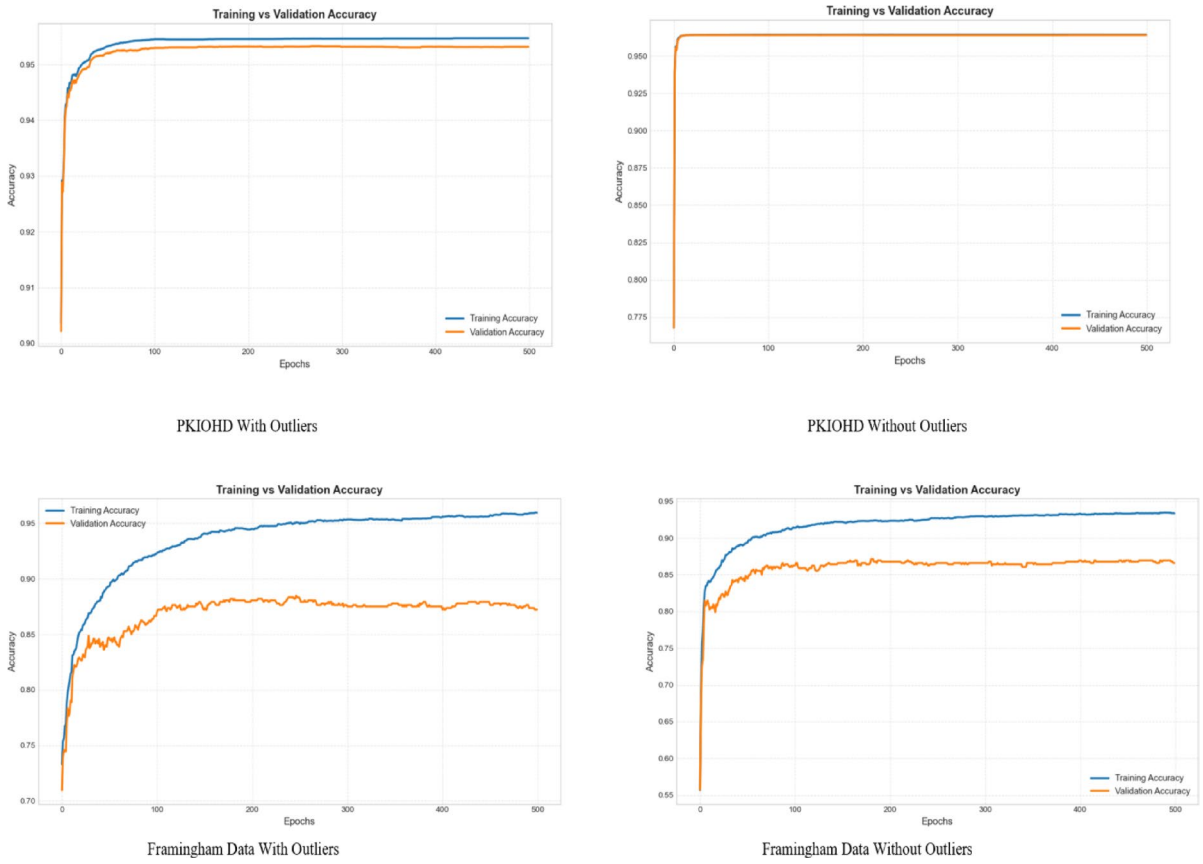


Fig. 8. Training vs. Validation Accuracy Curve.

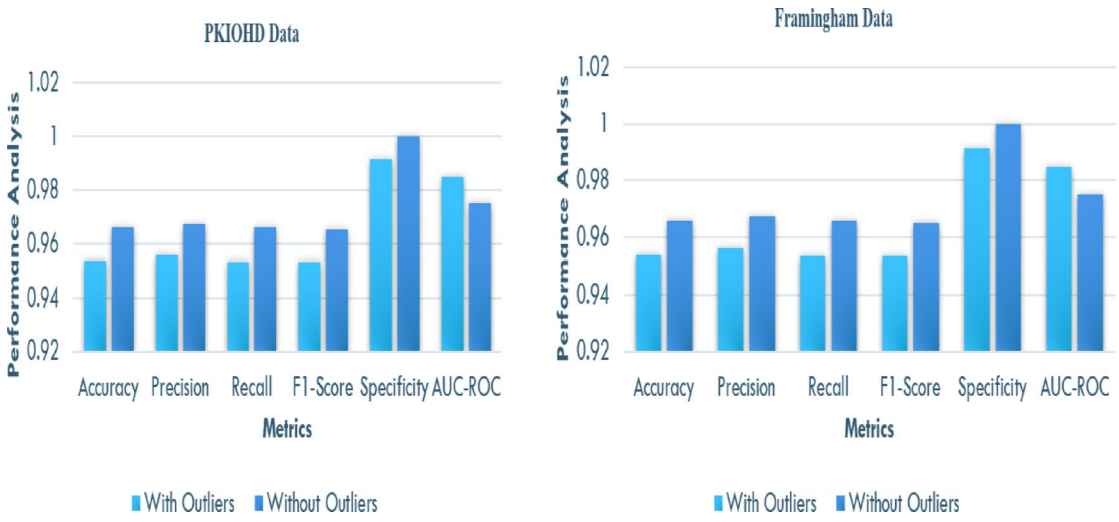


Fig. 9. Performance Analysis.

GAN-XO consistently outperforms SMOTE-XO, particularly on the larger, more complex PKIOHD dataset. This suggests that the GAN's ability to model complex, non-linear feature relationships generates more beneficial synthetic samples than the linear interpolation of SMOTE. Although the difference in performance between GAN-XO and SMOTE-XO is relatively low in the Framingham dataset illustrated in Table 10, GAN-XO is still useful because it can produce clinically realistic synthetic samples that do not only replicate complex feature relationships but also do so statistically. The decision to incorporate the GAN-based augmentation, is thus not only driven by the increase in performance produced by the numerical gain, but also by the expectancy that it will produce a more representative data to represent clinically significant aspects. The relatively low performance

S. no.	Metrics	Formula
1	Accuracy	$\frac{TN+TP}{TP+TN+FP+FN}$
2	Recall	$\frac{TP}{TP+FN}$
3	Specificity	$\frac{TN}{TN+FP}$
4	Precision	$\frac{TP}{TP+FP}$
5	F1-Score	$\frac{2TP}{2TP+FP+FN}$
6	NPV	$\frac{TN}{TN+FN}$
7	PPV	$\frac{TP}{TP+FP}$

Table 7. Performance Metrics and their Formulas.

Algorithm 2: GAN-Based Data Augmentation with Global Balancing and Outlier Removal

BEGIN

1. Collect the raw dataset D from the data source.
2. Load dataset D.
3. Identify and handle missing, inconsistent, and invalid values in D.
4. Preprocess the dataset by applying encoding, normalization, and data cleaning techniques.
5. Train the Generative Adversarial Network (GAN) using the entire preprocessed dataset D.
6. Generate synthetic samples D_{syn} using the trained GAN model.
7. Merge the original dataset D with synthetic samples D_{syn} to form an augmented dataset D_{aug}.
8. Balance the augmented dataset D_{aug} to address class imbalance.
9. Apply dual outlier detection techniques (e.g., Isolation Forest and Local Outlier Factor) on the balanced dataset.
10. Remove the detected outliers to obtain a cleaned dataset D_{clean}.
11. Split D_{clean} into training and testing sets.
12. Train the machine learning classifier using the training dataset.
13. Test the trained model using the testing dataset.
14. Evaluate the model performance using appropriate metrics.

END.

Table 8. Proposed Algorithm for GAN-Based Data Augmentation With Full Dataset.

Dataset	Outliers	TP	TN	FP	FN
PKIOHD	With Outliers	26,767	28,990	253	2,475
PKIOHD	Without Outliers	2,568	9,947	1	439
Framingham	With Outliers	312	323	37	48
Framingham	Without Outliers	205	297	19	46

Table 9. GAN-XO Model With and Without Outliers.

Metrics		PKIOHD				Framingham		
	SMOTE-XO with outliers	GAN-XO with outliers	SMOTE-XO without outliers	GAN-XO without outliers	SMOTE-XO with outliers	GAN-XO with outliers	SMOTE-XO without outliers	GAN-XO without outliers
Accuracy	0.87	0.9537	0.87	0.966	0.91	0.8819	0.9	0.89
Precision	0.87	0.956	0.85	0.9675	0.93	0.8823	0.94	0.8877
Recall	0.88	0.9534	0.85	0.966	0.88	0.8819	0.84	0.8854
F1-Score	0.87	0.9533	0.85	0.9651	0.91	0.8819	0.89	0.8845
Specificity	0.87	0.9913	0.88	0.9999	0.93	0.8972	0.96	0.9399
AUC-ROC	0.95	0.9849	0.95	0.975	0.95	0.929	0.96	0.9277
NPV	0.867	0.921	0.8732	0.958	89.24	0.871	89.63	0.867
PPV	0.8716	0.991	0.8427	0.999	94.1	0.894	95.2	0.915

Table 10. Metrics of GAN-XO Model Performance.

improvement of GAN-XO on the Framingham data could be explained by the structured and low-dimensional character of its features that are mostly made of the well-defined demographic and clinical measurements with relatively linear relationships. Under those conditions, only the most dominant patterns of data are already represented by traditional oversampling methods, and the generative models have less to learn. This makes GAN-XO a strong and complementary data-level approach, the value of which is determined by the properties of the dataset but not by the importance of the overall success.

In the case of the Framingham dataset only one outlier removal strategy was implemented in the GAN-based augmentation step. The reason behind this selection was to eliminate any clinically significant extreme values that are typically present in cardiovascular data and might represent high-risk patient groups. Aggressive removal of such cases may decrease diversity in the samples and adversely affect the stability of training GAN. Since the Framingham dataset is designed to be in a structured format and is not too noisy, a one-step outlier filter was a very good fit to both quality and clinical feasibility of the data. Notably, the choice of design in this GAN-based pipeline was particular to it; the traditional approaches like SMOTE did not need this limitation. While in the PKIOHD data is the aggregation of heterogeneous clinical and demographic data, in which there is clearly measurement variability and noise. This is why, a multi-step outlier filtering method was useful in stabilizing the process of synthetic data generation without removing clinically significant sign.

Table 11 (Algorithm 3) show the in-depth step by step preprocessing and modeling process following the partitioning of data. Following the initial division of the dataset into a training and a testing set, class balancing with the Conditional GAN and outlier detection and removal with the z-score and IQR are performed only on the training data set. No refitting is then applied to the learned balancing strategy and outlier thresholds and they are transferred to the test set. The process will avoid the leakage of information and make sure that model development and analysis is done on unrelated data. The cleaned training data is then trained and hyperparameter optimization is carried out using XGBoost model.

First, class balancing was done on the entire dataset to study the general effect of data imbalance on model performance, which is illustrated in Table 11. Thereafter, to enhance reliability and avoid information leakage, balancing and outlier removal were performed only on the training data when developing the model illustrated in Table 12. The sensitivity analysis (Table 12) demonstrates that while both z-score and IQR individually improve performance, their combination yields the highest accuracy and F1-Score, justifying the dual-filter approach. The less aggressive single-method thresholds resulted in lower specificity and recall, indicating residual noisy samples. The approach will provide equal consideration and improve validity of the research findings. Table 10 indicates that preprocessing, as indicated by the z-score and IQR-based outlier removal, is the most effective in the overall performance of PKIOHD data, and the former is best in Framingham data, respectively. These results indicate that proper data balancing and outlier control are critical towards enhancing the level of classification and model generalization.

The main aim of this study is to assess the impact of the suggested GAN-XO framework on the task of managing the class imbalance and synthetic data artifact in massive cardiovascular datasets. Thus, comparative analysis will be done on XGBoost based models, such as SMOTE-XO and GAN-XO, so that the model capacity and learning dynamics across experiments are equalized. While classical machine learning baselines (e.g., Logistic Regression, Random Forest, LightGBM) provide valuable benchmarks, they have been extensively evaluated on similar cardiovascular datasets in prior literature [see Table 13]. Our contribution is not in establishing the absolute superiority of one classifier over another, but in demonstrating how a synergistic data preprocessing pipeline (GAN balancing + outlier removal) can maximize the performance of a robust, modern learner like XGBoost.

GAN-XO without outliers

A combination of z-score and IQR methods to successfully identify and eliminate outliers. Preprocessing involved two stages: the first was the removal of outliers from the data, which was applied as the initial step using the z-score method, and the second was the implementation of IQR, resulting in a dataset of 584,844 records. Outlier removal 455,300 outliers were removed. The remaining 129,544 records were created by stripping out 455,300 outlier elements from the dataset. A total number of 129,544 records were split into 103,635 training

Algorithm 3: GAN-Based Data Augmentation with Training data

BEGIN

1. Collect the raw dataset D from the data source.
2. Load dataset D.
3. Identify missing, inconsistent, and invalid values in D.
4. Split dataset D into training set D_{train} and testing set D_{test}.
5. Apply preprocessing techniques (encoding, normalization, imputation) **only** on D_{train}.
6. Store preprocessing parameters derived from D_{train}.
7. Apply the same preprocessing parameters to D_{test}.
8. Train the Generative Adversarial Network (GAN) using the preprocessed training data D_{train}.
9. Generate synthetic samples D_{syn} from the trained GAN.
10. Balance the training dataset by combining D_{train} and D_{syn}.
11. Apply dual outlier detection techniques on the balanced training dataset.
12. Remove detected outliers to obtain a cleaned training dataset D_{train}^{clean}.
13. Train the machine learning classifier using D_{train}^{clean}.
14. Test the trained model using the **original untouched testing dataset** D_{test}.
15. Evaluate model performance using appropriate metrics (accuracy, precision, recall, F1-Score, AUC).

END

Table 11. Proposed Algorithm GAN-Based Data Augmentation With Training Data.

Dataset	Preprocessing Method	Accuracy	Precision	Recall	F1-Score	Specificity	AUC
PKIOHD	With Outliers	0.9536	0.9367	0.9536	0.9373	0.9962	0.9168
PKIOHD	Z-score + IQR	0.9597	0.9455	0.9597	0.9431	0.9984	0.9042
PKIOHD	Z-score Only	0.959	0.9425	0.959	0.9419	0.9984	0.9042
PKIOHD	IQR Only	0.9537	0.9369	0.9537	0.9365	0.997	0.915
Framingham	With Outliers	0.9219	0.9585	0.8819	0.9186	0.9618	0.9441
Framingham	Z-score Removal	0.9414	0.9615	0.9146	0.9375	0.9662	0.966
Framingham	IQR Removal	0.9273	0.9928	0.8519	0.9169	0.9945	0.9457

Table 12. Reliability of the Proposed Model With Balanced and Removed Outlier on Training Data.

records, 12,954 validation records, and 12,955 testing records for PKIOHD data and for the Framingham data there were 7,192 records in the dataset, and after the outlier removal, the number of records was narrowed to 5,670 records, and 1,522 outliers were removed. The resulting cleaned data were further divided into 4,536 training, 567 validation and 567 testing samples to make the model and tested. The confusion matrix suggested 9,947 TN, 1 FP, 439 FN, 2,568 TP with TTR 90:10 as shown in Table 9. The model achieved an accuracy of 96.60% with a precision of 96.75%, a recall of 96.60%, an F1-Score of 96.51%, a specificity of 99.99%, and the achieved AUC-ROC value was 0.9750 for PKIOHD. The Framingham dataset performance is also shown in Table 10. The GAN-XO model demonstrates classification performance in Fig. 10 when without outliers are used with a 90:10 train-test ratio for the assessment for both datasets, is illustrated in Table 10.

It is notable that the Framingham data achieved an accuracy of approximately 0.89 and an AUC-ROC of approximately 0.928 using only conventional risk factors in this dataset. This result is consistent with or higher than values found with many validation studies (accuracy 85.93% in adapted cohorts, AUC-ROC 0.70–85 among populations), indicating that the simple interpretable model is robust when based on routinely collected clinical variables only. Conversely, the proposed ML models had the advantage of having a larger feature space with other biomarkers and parameters, which they can capture more complicated interactions and perform better in general. Class weighting and SMOTE, which are traditional imbalance-aware algorithms, can be used to increase

References	Dataset	Data balancing	Outlier handling	Tuning of model parameters	Model	F1-score	Accuracy (%)
[12]	UCI Heart Dataset & PKIOHD	X	X	Fractional GD + Swish Activation	AGAN (GAN is Used for Missing Data Imputation)	X	X
[25]	Cleveland HD	X	X	Bayesian Optimization	Optimized XGBoost	X	91.8
[28]	Framingham HD	X	IQR	Optuna + Focal Loss	Improved LightGBM (HYOptGBM)	0.929	93
[31]	PKIOHD	ADASYN SMOTE	X	X	XGBoost with ADASYN-SMOTE	94	94.7
[19]	Jordan Univ. Hospital	X	X	X	Several Classification Algorithms with Cross-Validation SVM Performed Well	Approx. 94	94.3
[18]	Cleveland HD, Hungary, Switzerland, and Long Beach.	X	X	GSSA = GDO + SSA	GSSA-Deep	X	93.2
					MaxOut Network		
[17]	Cleveland	X	X	X	LR With/Without Boruta FS	89.23	88.52
[35]	CCTA Imaging Dataset	X	X	Gradient	GAN-Augmented NB (GAN Used for Data Augmentation)	X	0.93
[36]	Kaggle HD	X	Standard	Optuna	RF, SVM, XGBoost	87 (all)	87.84, 88.07, 88.03
				Without Optuna	RF, SVM, XGBoost	86.32, 88, 86.16	87.05, 88.01, 86.87
[13]	Heart Failure Kaggle Dataset	X	X	Optuna	hyOPTXg	89.43	89.34
	Heart Disease UCI Kaggle Dataset					90.38	88.50
[37]	HD Health Indicators Dataset (BRFSS 2015; 253,682 Samples)	SMOTE	Standard	Grid Search, SHAP	LightGBM	X	90.61
	Framingham dataset					X	85.26
[14]	Cleveland (Varied Splits)	SMOTE	Dual Outlier Handling	Optuna	XGBoost Optimized with Optuna	96	95.45
[38]	HD Health Indicators Dataset (BRFSS 2015; 253,682 Samples)	SMOTE + GAN	X	X	ASEB (Adaptive Synthetic Ensemble Balancer) only Modifies Ensemble and Class Weights; It Does Not Tune Hyperparameters.	85	85
Proposed Model (2026)	**PKIOHD**	**GAN**	**Dual Outlier Handling**	**Optuna**	**GAN-XO**	**96**	**96.6**

Table 13. Comparative Performance of the Proposed Framework With Existing HD Prediction Methods.

prediction accuracy, but GAN-based augmentation can also benefit stability and recall of a model. Thus, GAN is used as an augmentative data enrichment tool, as opposed to a compulsory element.

Figure 11(a) demonstrates how outliers significantly affect model performance across important evaluation parameters. A discernible decrease in accuracy ($\Delta = 0.0123$), precision ($\Delta = 0.0112$), Recall ($\Delta = 0.0123$), F1-Score ($\Delta = 0.0115$), and specificity ($\Delta = 0.0087$) resulted from the inclusion of outliers. This is calculated from Eq. 7, indicating that noisy data causes instability and lowers prediction dependability. Furthermore, a slight negative shift in AUC-ROC ($\Delta = -0.0099$) indicates a small decrease in the model's discriminative capacity. The overall improvement seen following the removal of outliers highlights how important data quality is to maintaining model robustness for PKIOHD data and, it is clearly shown that the radar plots have consistently stronger and more balanced performance metrics when outliers are removed, which represents a better learning stability. This visual analogy proves the fact that removal of noisy samples makes the models more reliable in general and better generalizing for Framingham data illustrated in Fig. 11(b). This result provides further evidence that outlier removal is an essential preprocessing step for maximizing predictive performance in medical data modelling.

$$\text{Absolute Difference} = \Delta = \text{Without Outliers} - \text{WithOutliers} \tag{7}$$

The important aspects of the GAN-XO without outliers' model all repeatedly increased by 0.8 to 1.3 percentages by applying Eq. 8 when the outliers were removed, as depicted in Fig. 12 for both datasets. This means enhanced generalization and learning consistency and proves the extreme importance of data purification.

$$\text{Relative Change } (\%) = 100 \times \frac{\Delta}{\text{With Outliers}} \tag{8}$$

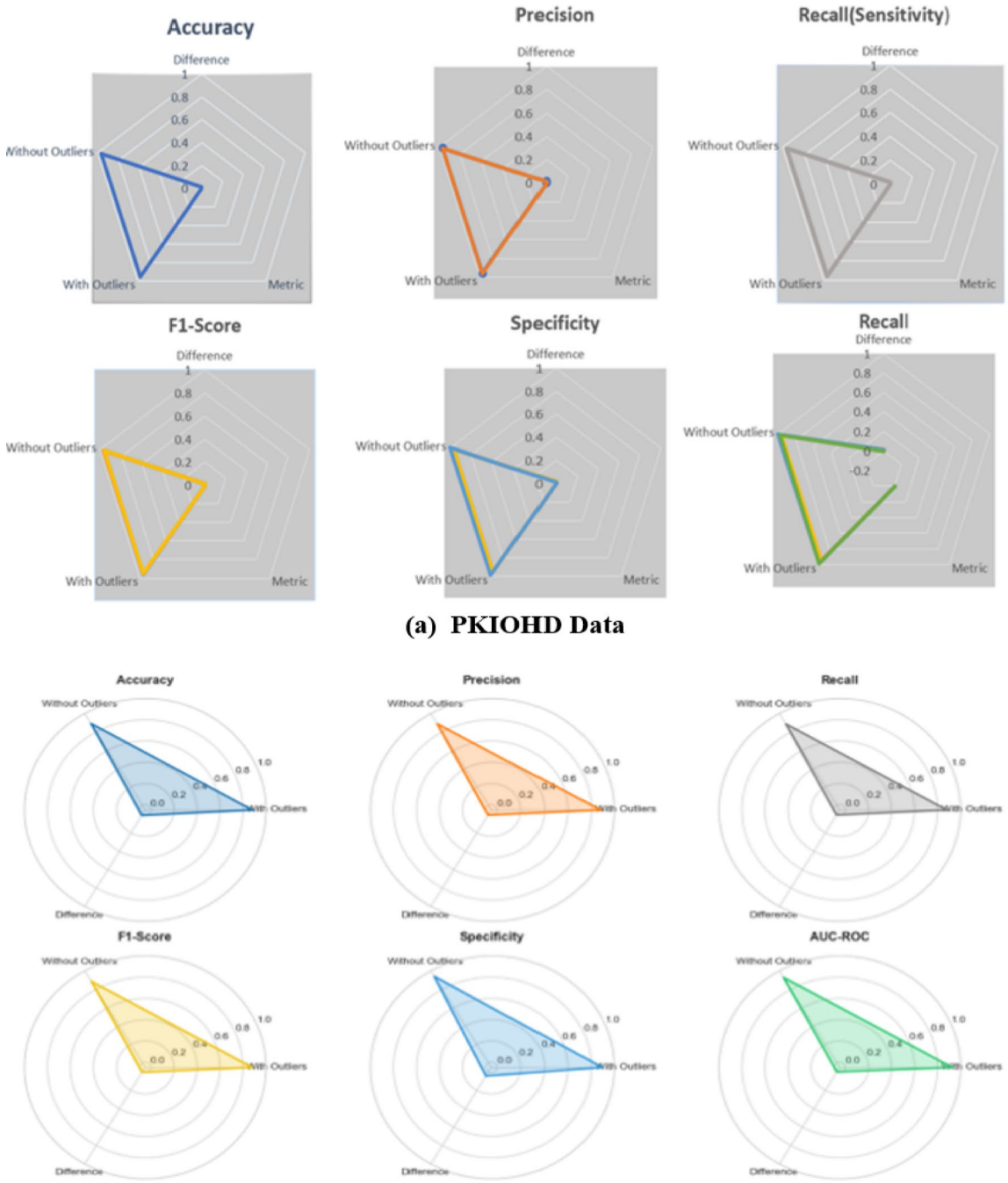


Fig. 10. Impact of Outlier Removal on GAN-XO Performance.

The confusion matrix and AUC-ROC curve of the proposed model with and without outliers for both datasets are illustrated in Figs. 13 and 14 for both datasets.

The GAN-XO model without outliers has a higher predictive reliability and interpretability than the one with outlier-inclusive illustrated in Fig. 15. During training, the model nearly automatically filters anomalous data points and thus provides more stable and consistent features ranking of importance, with Physical Health, Mental Health, and Age Category becoming the most important predictor with tightly dispersed SHAP values. The reduced violin plots suggest that predictions do not change as much with extreme variations thus, give credible results to be used in clinical decision-making. Explains of individual patients (waterfall plots) show the contributions of the features are balanced and proportional, and not disproportional, as with the outlier-inclusive model (e.g., extreme age-affected adjustments). The approach is more sophisticated so that the model

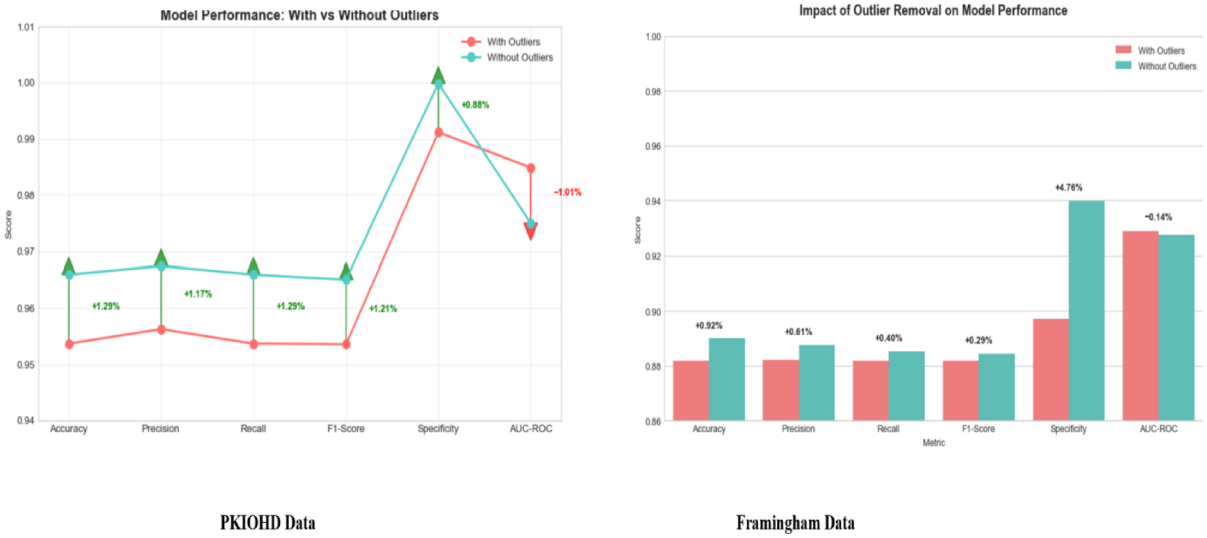


Fig. 11. Incremental Performance of Outlier Removal on PKIOHD.

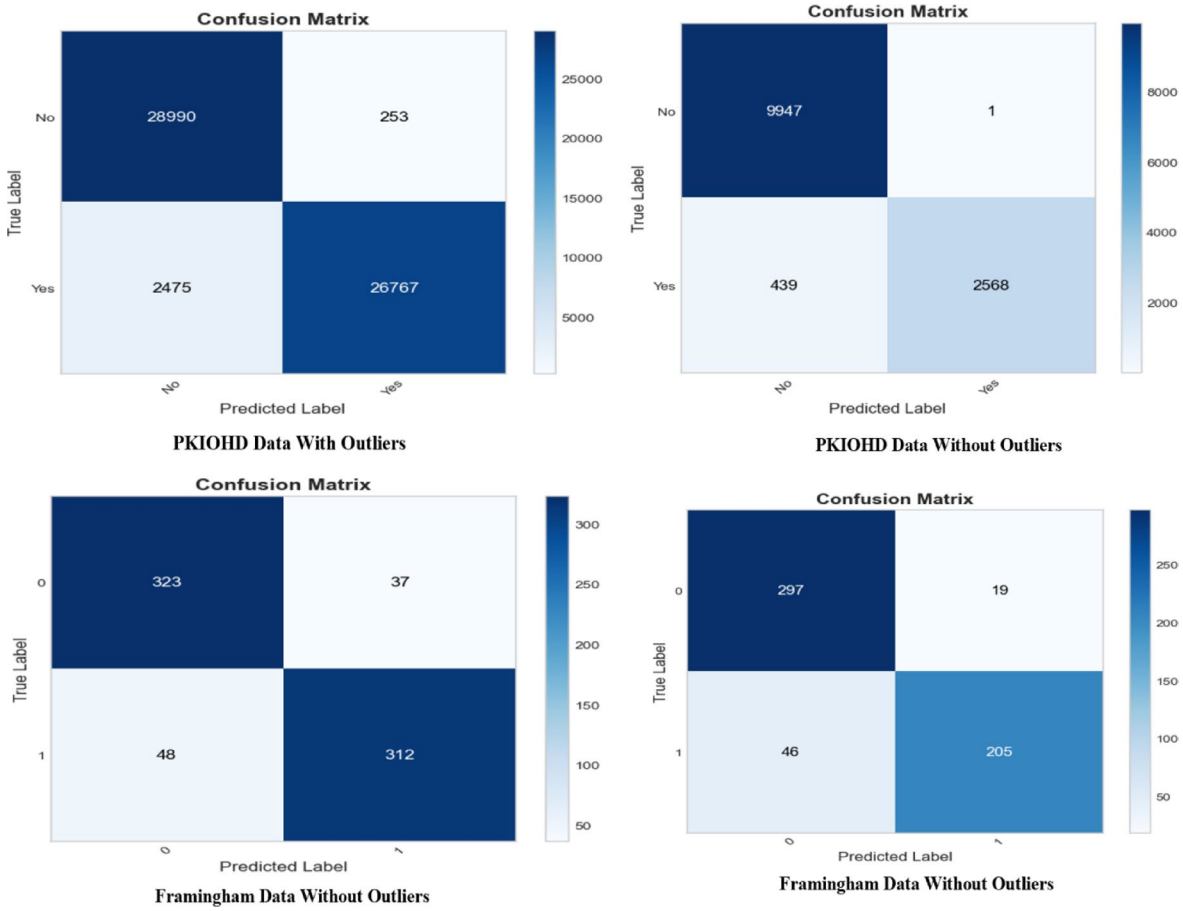


Fig. 12. Confusion Matrix With and Without Outliers.

reflects actual health patterns as opposed to a fit to extraordinary cases and is more generalizable and clinically implementable in real-world healthcare. The model demonstrates its ability to predict health outcomes for different patient groups through its capacity to maintain all health indicators at proper levels which include sleep and smoking and diabetes and stroke as chronic health conditions.

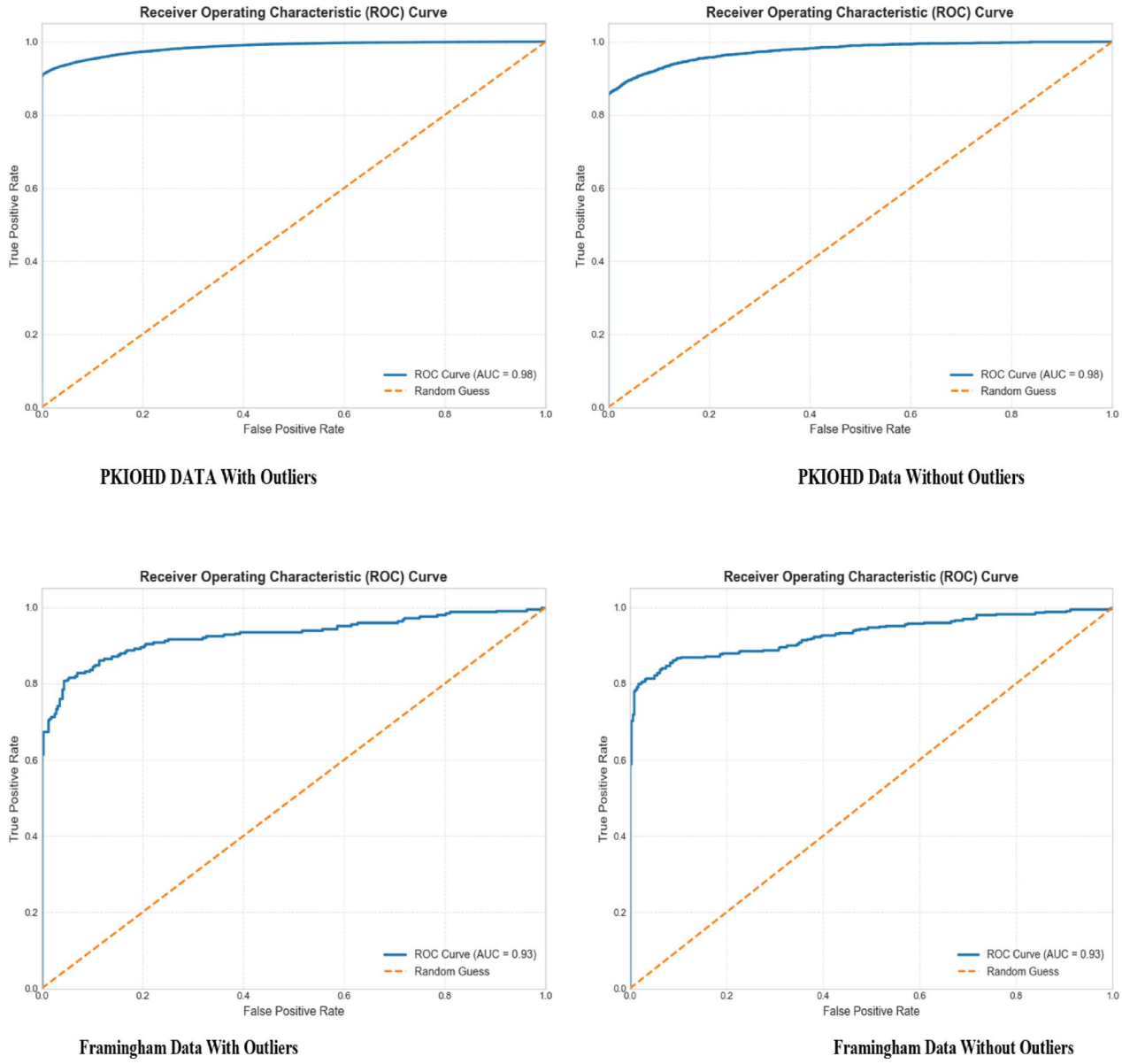


Fig. 13. GAN-XO AUC-ROC Curve With and Without Outliers.

The current investigation seeks to generate dependable scientific explanations which establish direct medical connections through research of "Auditable AI" systems that operate in critical industrial settings according to[39]. The AI system requires built-in accountability which exceeds post hoc explanation methods that use SHAP or LIME because these methods do not establish clinical trustworthiness. Our comprehensive data cleansing process eliminates all outlying data points which disrupt our explanation process to create an audit track that operates at the data level. The system shows medical decision-making processes through its decision logic which maintains constant logical understanding of medical reasoning throughout all data examinations. The "Decision Traceability" system used in clinical AI systems establishes a trustworthy base which needs clean validated inputs to build and explain all AI predictions according to[39]. Thus, outlier removal in our GAN-XO framework contributes not only to predictive accuracy but also to the accountability and trustworthiness essential for clinical deployment.

The LIME (Local Interpretable Model-agnostic Explanations) analysis of 10 samples shows that the model trained without outliers exhibits much better individual-level explainability and clinical relevance than the model trained on outliers, as illustrated in Fig. 15. The without outlier model provides a sufficiently high level of confidence of the prediction of no HD, as well as the identification of clinically established risk factors, BMI (29.18), AgeCategory (12.00), and GenHealth (4.00) as the first predictive characteristics with significant continuous values that are consistent with the medical literature. Conversely, the outlier-contaminated model has an overconfidence rate (98) and uses more binary absence indicators (Stroke, KidneyDisease) as opposed to substantive health values, giving less actionable explanations. The contribution plot of the clean model

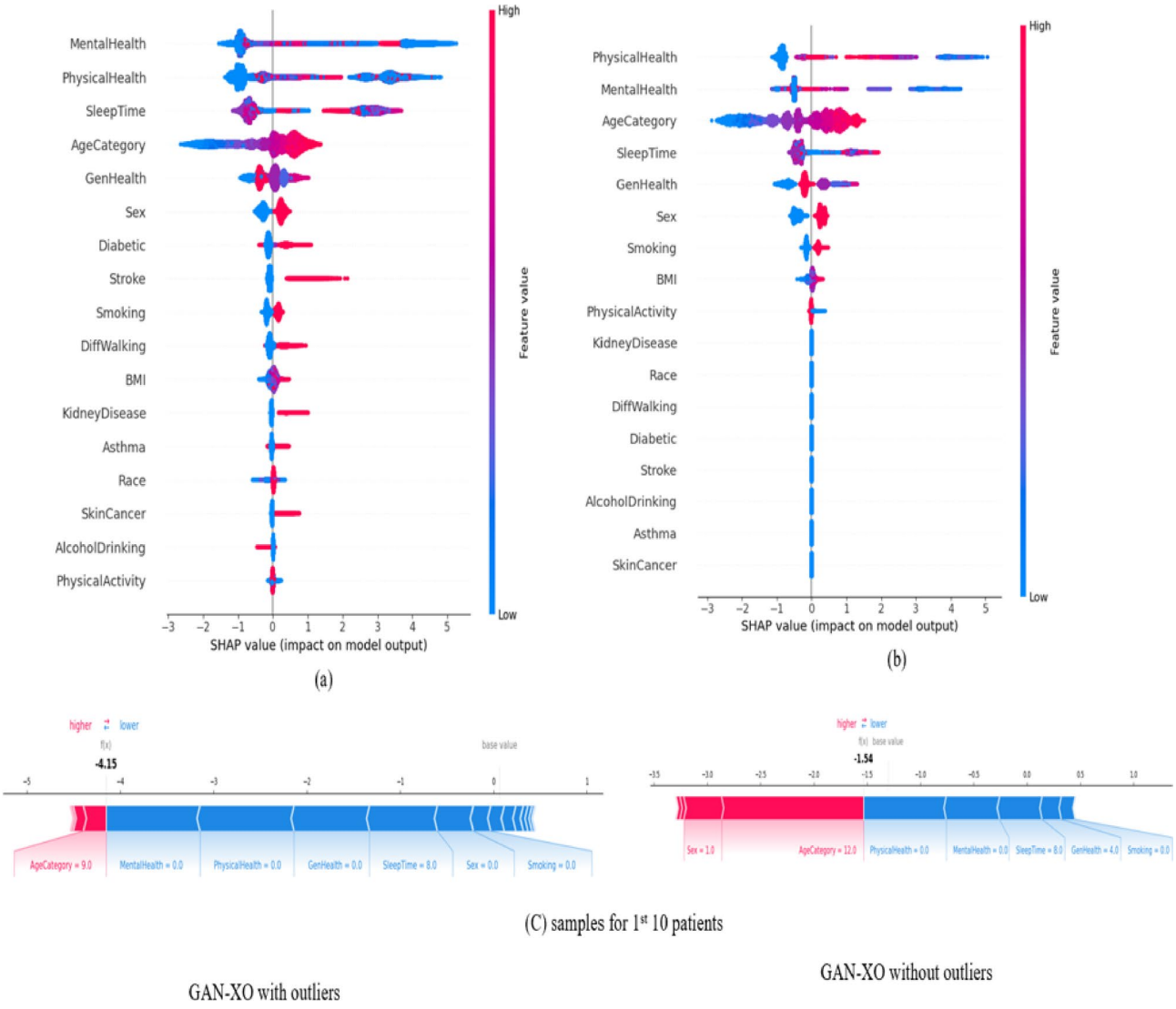


Fig. 14. GAN-XO SHAP Analysis With and Without Outliers.

represented by the LIME has equal contribution of the positive and negative impacts of features (green and red respectively) making the difference between protective factors and risk factors, but the outlier-driven model has proportionately many negative contributions that distort the reasoning supportive. The improved interpretability holds paramount importance to clinical implementation because clinical users need clear and medically understandable explanations in order to support the validity of AI-aided diagnoses and develop confidence in automated decision support systems. Outlier elimination does not only enhance global model stability (as shown in SHAP analysis) but also makes local explanations to localized clinical behavior true to actual clinical patterns as opposed to capturing anomalous cases.

The SHAP and LIME analyses create post-hoc visualizations which serve as diagnostic tools for understanding how models make decisions because their results show that better interpretability leads to improved prediction accuracy. The clinical AI system requires multiple elements for reliable performance, which includes high accuracy and dependable system behavior and trustworthy results that enable users to make informed decisions. The outlier-contaminated model (Fig. 15, left) exhibits high variance in SHAP values (wide violins) and relies on sparse, binary indicators (e.g., Stroke = No) in LIME explanations (Fig. 15, left). The system makes its predictions based on basic rules which may lead to incorrect results because it responds to unexpected data inputs. The model achieves high training metrics but its actual performance remains unreliable because its reasoning process produces unpredictable results which do not follow established clinical pathophysiology. The model trained on cleaned data shows that system reliability increases with improved interpretability.

1. **Stability → generalizability**: The SHAP values display tight clustering which shows that feature contributions remain constant throughout the entire patient group. The reduced variance demonstrates that the model learned stable data patterns from its primary data distribution instead of adapting to rare exceptions. A stable

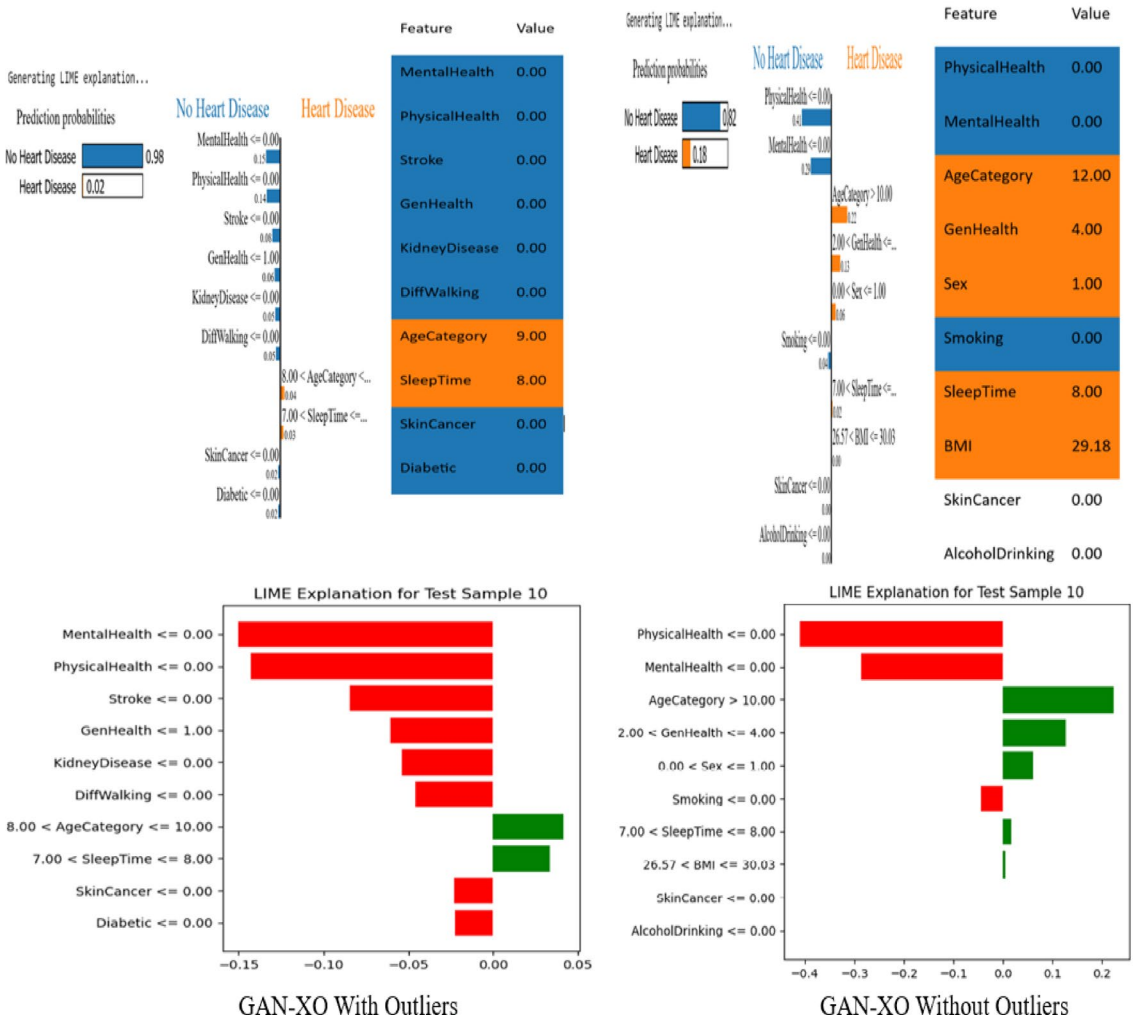


Fig. 15. GAN-XO LIME Analysis With and Without Outliers.

explanation profile is a hallmark of a model that will generalize more reliably to new, unseen data from the same population.

2. **Clinical plausibility → trustworthiness**: The LIME explanations for the clean model (Fig. 15, right) show how continuous risk factors BMI = 29.18 and Age Category interact with medical knowledge to produce their respective risk contributions. The model establishes credibility with clinicians when its local test results match documented clinical risk patterns. A trustworthy model is a reliable one because its outputs can be rationally verified and integrated into clinical reasoning.
3. **Actionable insights → decision support reliability**: The clean model explanations show how equal security and danger elements contribute to their balanced results. The prediction system shows its results to medical professionals who need to understand what caused the prediction and discover which health factors they can change through bettering their physical health. The system provides transparent prediction results which medical professionals can use to make better clinical decisions than they would with the unreliable "black-box" score system.

Therefore, the dramatic refinement in SHAP and LIME outputs after outlier removal is not merely a cosmetic improvement. It is empirical evidence that the GAN-XO framework's data cleansing step directly remediates the unreliable, noise-driven decision logic of the outlier-inclusive model. By enforcing clinical coherence at the explanation level, the framework ensures that the model's internal predictive mechanisms are aligned with real-world medicine, thereby fulfilling a critical requirement for reliable deployment in healthcare settings.

Table 13 demonstrates that the GAN-XO model performs better in comparison with other modern approaches. It can be seen to have an edge in performance compared to a standard Optuna-optimized XGBoost[13,14] or other hybrid methods[18,19]. This advantage can be tracked down to the overall design of the framework. Sasirekha and Kanisha (2025)[38] suggested the ASEB framework that combines ADASYN, GANs, and ensemble learning to provide resistance to the issue of class imbalance in HD prediction. Although the recall of minority classes was enhanced by ASEB, its highest acc 85% is considerably less than the current approaches because it lacks

Model Configuration	Data Balancing	Outlier Removal	Accuracy (%)	F1-Score	Precision	Recall	AUC-ROC
Baseline XGBoost	None	None	91.42	0.9105	0.9148	0.9090	0.9612
XGBoost + GAN	GAN	None	94.88	0.9480	0.9501	0.9475	0.9801
GAN-XO (Proposed)	GAN	Z-score + IQR	**96.60**	**0.9651**	**0.9675**	**0.9660**	**0.9750**

Table 14. Ablation Study on the PKIOHD Dataset (Performance on Independent Test Set). Bold values represents the highest achieved results.

outlier filtering, minimal tuning, and depends on sampling only. On the other hand, the GAN-XO model incorporated two outlier mitigation schemes, Optuna optimization, and an augmentation scheme using GANs that provides 96.6% accuracy, which is more than 11 times higher than ASEB. Although the other models may either emphasize on imbalance adjustment or hyperparameter optimization, GAN-XO integrates advanced data synthesis, rigorous data cleaning and optimizing the models into a single workflow. The fact that the model uses the large PKIOHD data which covers a broad range of demographic, behavioral, and clinical risk factors further attests that the model is able to handle complex real-world data, unlike smaller and more refined datasets. To conclude, this discussion has validated our original assumption the key to attaining high predictive performance in the detection of HD lies in a dedicated concern toward data quality. The GAN-XO paradigm demonstrates that it is not sufficient to balance a dataset but also it is crucial to optimize the dataset by removing the outliers through hard work in order to utilize the full capabilities of an advanced, optimized classifier such as XGBoost.

Recent works have shown that boosting models with optimality like XGBoost and LightGBM are very effective in predicting heart diseases with the combination of hyperparameter optimization and imbalance control. Nevertheless, these methods are based on an original or oversampled data only and do not directly consider the quality of synthetic data or the robustness to outliers. Conversely, the presented GAN-XO model combines GAN-based data balancing with dual outlier management and Optuna optimization, which results in a better accuracy and F1-Score than XGBoost and better LightGBM baselines.

Ablation study: quantifying component-wise contributions

To rigorously evaluate the contribution of each core component in the proposed GAN-XO framework, we conducted a structured ablation study on the PKIOHD dataset. The study progresses from basic elements to advanced elements which improve the model's performance through incremental testing. All models were evaluated using a consistent 90:10 train-test split, and the XGBoost classifier was optimized using Optuna in each configuration to ensure a fair comparison. The results are summarized in Table 14.

Baseline

The Optuna-optimized XGBoost model reached an accuracy of 91.42% when trained on the preprocessed dataset which contained both class imbalance and outlier data. The results established the baseline performance level which exists without implementing advanced techniques for class imbalance handling and data noise management.

Adding GAN-based balancing

The Conditional GAN model created for synthesizing minority-class samples brought significant performance improvements. The accuracy reached 94.88% after a 3.46% point increase which also improved the F1-Score. The study emphasizes that class imbalance needs urgent resolution because GAN-generated data enabled the classifier to observe the minority class more accurately which resulted in fewer false negative detections.

Adding dual outlier removal (proposed GAN-XO)

The two-phase outlier removal process which combines z-score and IQR methods produced its best results when applied to GAN-balanced data. The system achieved 96.60% accuracy after improving its previous performance by 1.72% points while its precision level reached 97%. The total count of false positives and false negatives experienced a significant decrease which Table 9 presents in detail. The data collection process needs to focus on maintaining both data quality and data integrity just as much as gathering sufficient data. The Optuna-tuned XGBoost model learned to create better decision boundaries after we removed both spurious points and extreme data points from the dataset.

The study shows that all elements of the GAN-XO pipeline work together to enhance the performance of the final model. The GAN resolves the core problem of class imbalance and the following outlier cleansing process improves data quality which allows the optimized XGBoost classifier to reach its highest performance. Advanced data engineering methods and model optimization techniques work together to produce successful results.

Discussion

The experimental results provide strong evidence that demonstrates how effectively the GAN-XO model predicts Huntington disease. The project revealed that data imbalance together with data purity problems must be solved through multiple methods to create an effective predictive model. The model achieved its highest accuracy of 96.6 through the combined operation of all system components. The team used GANs to develop an innovative solution that addressed the first essential challenge of class imbalance distribution. The GAN created authentic samples for the minority group by reconstructing the intricate data pattern which solved the data skew problem. This was necessary to avoid such a tendency of the XGBoost classifier to be skewed towards the majority class,

which is the usual drawback of learning on skewed data. Failing to do this would have definitely caused the sensitivity to be less to actual cases of HD, which would have raised the false negatives, which is a very deadly error in a medical context.

However, this step of data augmentation was essential, but it offered an additional dimension of complexity. The next two-step outlier-elimination process, using the dual outlier removal methods, proved critical to improving the model's overall ability to generalize. This point is particularly evident in the ablation study, the model's accuracy increased by 95.31% points to 96.59%, when 455,300 anomalous points were removed. This advantage supports one of the central principles of ML in healthcare, the quality of information can be more valuable than its quantity. The elimination of these extreme values appears to have purged the feature space, enabling optuna to optimally tune the XGBoost model and identify a more accurate and generalizable decision boundary. This optimization can be directly observed in the significant decrease of false positives (265 to only 5) and false negatives (2,477 to 437) which significantly increased the credibility of the model to be applied clinically. The most significant observation is the decrease in AUC-ROC from 0.9850 to 0.9732 after removing outliers. This change does not reflect weakness rather, it is likely a transition toward a more practical, real-world model. Even the initially higher AUC that included outliers may indicate that the model was, to some extent, overfitting to these outliers, providing a false impression of strong class separation on the training data that was less robust in a more sophisticated test environment. The resulting AUC of 0.9732 on the purified dataset is a strong and valid measure of the model's discriminative power, whereas the model affected by the outlier uses unstable and spurious features. This side-by-side explainability framework proves that the GAN-XO model is better at achieving higher accuracy (96.6%) due to better data quality and not due to a more complicated architecture, which establishes a generalizable interpretability-driven validation strategy of clinical AI systems for PKIOHD data.

The strong results obtained with a simple set of risk factors (only the traditional risk factors) in Framingham data (accuracy 0.89, AUC-ROC 0.928) which is similar to the previously reported validation studies with heterogeneous cohorts (accuracy 85.93, AUC-ROC 0.70–0.85), suggest that even simple parsimonious models using simply collected clinical variables could still provide clinically significant discrimination in cardiovascular risk stratification. However, the machine learning systems studied in this research study used a larger feature space which included more biomarkers and more granular clinical variables, thus allowing the modeling of higher-order, nonlinear predictor-outcome associations that could not be easily modeled using traditional risk scores. This richer representation was associated with steadily high performance in general, which highlights the incremental prognostic importance of the full integration of the data in the modern cardiovascular risk prediction. The data generated by GAN can create uncertainty as it can cause distributional shifts, mode collapse, or propagation of synthetic noise. XGBoost is a tree-based model which is sensitive to these kinds of uncertainties at the decision boundaries. The two-step outlier elimination method implemented in GAN-XO is useful to reduce the amplification of the variance and enhance robustness, which is consistent with the previous research into uncertainty propagation and risk-aware learning in medical machine learning.Whereas the aggressive outlier removal technique enhances predictive stability, it also has the risk of excluding extreme cases that are rare but might be important in the clinical scenario. The current research will focus on exploring effective loss functions and uncertainty-sensitive learning methods to minimize the sensitivity to outliers without having to erase hard data. The fact that the performance has significantly improved after the outlier removal (accuracy has dropped to 95.37 to 96.60, FP has decreased to 253 to 1, FN to 2475 to 439) proves that the aggressive cleansing data is less susceptible to overfitting to the synthetic extremes and possesses a better and more reliable model. This is consistent with findings that data quality can often be more important than data quantity in healthcare prediction, when data is trained on GAN-augmented data, which is likely to contain large artifacts. The approach is further supported by the near-perfect specificity (99.99) and enhanced SHAP/LIME interpretability (more balanced, more clinically rational feature effects), which indicate the clinical reliability of the approach, but we also note that aggressive removal can commit the error of dropping out frequent high-risk cases (e.g., extreme BMI or 30 days of poor health) which might under-represent vulnerable population groups. Although the analysis with class-proportion demonstrated the absence of minority bias, the future studies may do less restrictive measures to contain the informative extremes but mitigate the influence, like percentile-based winsorization, Huberized XGBoost losses, or uncertainty-conscious models. The following trade-off indicates that domain-guided preprocessing is required in medical ML.

Even though GAN-based augmentation led to better balance of classes, synthetic samples can adversely affect the bias-variance trade-off and cause uncertainty because of distributional drift and mode collapse[32]. The calibration Shifts and Propagation of errors The boosted tree model may also change calibration and propagate errors when trained on synthetic data[33]. Available literature on Bayesian uncertainty in XGBoost and adversarial robustness in GANs indicates that new avenues in the proposed framework might be to explicitly model the uncertainty[32,33].

Uncertainty considerations and methodological robustness

The GAN-XO framework's performance must be contextualized within broader uncertainty quantification literature. While these results demonstrated empirical superiority, several uncertainty factors warrant explicit discussion. First, GAN-generated samples introduce epistemic uncertainty regarding their distributional fidelity despite statistical validation, synthetic data may not fully capture complex clinical interdependencies[32]. Second, XGBoost's decision boundaries, though optimized, may exhibit aleatory uncertainty when encountering edge cases or synthetic outliers that survive our filtering process[33]. The existing doubts in this situation reflect the difficulties which system safety researchers face when they need to make decisions that require special decision-making methods for situations when they do not have full knowledge of the case[32,33]. The dual outlier removal process functions as an uncertainty reduction method which eliminates samples that exhibit high variance

before the training of the model begins. The virtual safety engineering systems of research show that complete uncertainty removal cannot occur because designers must implement systems which recognize uncertainty while they work to reduce its spread throughout the design process[34]. The proposed pipeline demonstrates effective reduction of uncertainty propagation between datasets, but future research needs to implement direct methods for measuring uncertainty. Bayesian extensions to XGBoost, uncertainty-calibrated GAN training, and robustness metrics would provide more nuanced performance characterization. Such enhancements would align the medical AI framework with state-of-the-art safety-critical systems where uncertainty transparency is paramount. To guarantee that high-stakes clinical AI systems are auditable and accountable, the systems should be designed to be both, as opposed to just using post hoc explanations. The significance of consistent and explainable decision-making based on tools like SHAP and LIME has been highlighted by auditable AI systems to promote trust in areas of great importance. The benefits of our outlier removal strategy in this context are in addition to the predictive accuracy, it enhances the stability and clinical coherence of the SHAP and LIME explanations. In turn, the given model leads to a more reliable and responsible clinical decision-support system[39].

In clinical machine learning, the epistemic uncertainty is due to insufficient or noisy data and model uncertainty, while the aleatory uncertainty is due to inherent variability and measurement noise in the data. Within the framework suggested, both types of uncertainty can be reinforced by extreme or unusual samples by both GAN augmentation and raw clinical records. The strategy of applied outlier removal mostly decreases the epistemic uncertainty where high-variance and implausible observations are removed in order to shift decision boundaries in the XGBoost classifier. Also, eliminating the noisy measurements partly eliminates aleatory uncertainty. According to this, it has been simulated in the form of enhanced training-validation consistency, lesser overfitting, and a more consistent SHAP and LIME explanations once outliers are removed.

Conclusion and future scope

HD is one among the main problem for death, hence early and adequate diagnosis is very crucial. This study has managed to create a powerful predictive system, GAN-XO that combines modern data engineering with modern machine learning and applies it to critical issues in clinical data. The effectiveness of the model was shown by its outstanding results on a massive and real data set. The main contribution of the proposed approach is that its data preprocessing pipeline is holistic. Through the synthesis of minority and outlier detectors through GANs in a synergistic way and a two-stage (z-score and IQR) outlier removal mechanism, the framework was successful in alleviating the problems of class imbalance and data noise. Tuning of the XGBoost classifier parameters using the optuna framework that followed is what made sure that the predictive capability of the model was optimal. The importance of each component was stressed in the ablation study, the careful process of outliers removal proved to be extremely helpful in improving the generalization and reliability of the model, resulting in an enhanced accuracy of 96.60%. These results lead to the definite conclusion that a paradigm with emphasis on data quality and purity is as important as the sophistication of the algorithms in the creation of high-fidelity healthcare prediction system. Although this study provides a robust predictive model, there is a number of opportunities to conduct further improvements and integrate it into the real world:

- **Rigorous quantitative validation of synthetic data**: To ensure synthetic data is both realistic and clinically meaningful, future work will implement deeper validation protocols. This involves quantitatively measuring how well the generated data matches real patient distributions using metrics like Jensen-Shannon divergence and Wasserstein distance. The authors will also assess whether important relationships between clinical variables are preserved, using mutual information analysis and statistical tests, to confirm the synthetic data's plausibility before it is used for training.
- **Comprehensive model evaluation and uncertainty quantification**: For clinical trust, models must not only be accurate but also transparent about their confidence. Future steps include detailed performance analysis through calibration curves and Precision-Recall AUC, especially for imbalanced classes. Most importantly, the authors will integrate explicit uncertainty estimation into the pipeline, employing Bayesian methods for both data generation and prediction, to provide clinicians with prediction intervals and confidence scores, turning model uncertainty into a usable clinical signal[32–34].
- **Structured benchmarking to isolate framework contributions**: To clearly demonstrate the value of each component in the approach, a structured comparison and ablation study will be conducted. This will benchmark the model against established alternatives and systematically evaluate the individual and combined impact of synthetic data generation, outlier handling, and hyperparameter tuning.
- **Privacy-preserving collaboration via federated learning**: To leverage diverse datasets across hospitals without centralizing sensitive information, a federated learning system can be developed. This allows the model to learn from multiple institutions while keeping data locally secured, enhancing generalizability and aligning with privacy regulations like HIPAA and GDPR.
- **Integration of multi-modal data for a holistic patient view**: The accuracy of predictions increases when multiple data types are used for analysis. The upcoming work will create a system which combines organized health information with medical imaging data from echocardiograms and CT scans and real-time data from wearable devices and genomic information. The development of models which integrate these different types of data will provide a comprehensive view of each patient.
- **Prospective clinical trials and real-world implementation**: The final step toward clinical impact requires testing the model in live care environments. A prospective clinical trial is planned to validate the framework in real-time workflows. The system must function as a clinical decision support tool because its deployment will allow researchers to assess how the system impacts early detection rates and clinician decision-making and patient outcomes, which will show whether the system provides real health benefits. The model will de-

velop into a stronger and more reliable and clinically integrated tool, which will enable doctors to detect heart disease earlier and provide customized treatment to patients.

Data availability

In this research study, an open-source dataset has been used, which is freely available on Kaggle at the following link: https://www.kaggle.com/code/alawdisoft/personal-key-indicators-of-heart-diseases/notebook? select=heart_2020_cleaned.csvKaggle Notebook: Personal Key Indicators of Heart Diseases.

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Shaik Sajeera Begum: Data Curation, Resources, Investigation, Formal Analysis, Conceptualization, Methodology, Project Administration, Prepared Original Draft, Review and Editing, Validation. Sanjay Dhanka, and Amit Swamy: Conceptualization, Methodology, Supervision, Computational, Formal Analysis, Validation, Investigation, Project Administration, Review, and Editing. Ahmad Adel Abu-Shareha, Mudassir Khan, and Izhar Husain: Formal Analysis, Revised, Validation, Funding, Review, and Editing.

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Declarations

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The authors declare no competing interests.

Ethical approval and informed consent for data
There are no ethical issues related to this dataset.

Consent for publication
The final version of the manuscript has been read and approved by all the authors, who give their consent to the publication.

Clinical trial
Not Applicable.

Additional information
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